LecSeg-30:

A Reproducible Low-Resource Benchmark and Diagnostic Study

for Lecture-Video Topic Segmentation

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Declaration

We hereby declare that this thesis is based on the results found by ourselves. Materials of work found by other researchers are mentioned by reference. This thesis, neither in whole nor in part, has been previously submitted for any degree.

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Abstract

Recorded lecture archives have grown faster than the infrastructure for navigating them. Platforms such as Coursera, edX, and YouTube collectively host millions of academic videos, yet a student searching for a specific derivation or argument within a two-hour recording has few options beyond manual scrubbing. This thesis asks how far a small, carefully constructed benchmark can take us toward automatic lecture topic segmentation when supervision is limited to thirty videos.

LecSeg-30 is the resulting benchmark: 30 publicly available YouTube lectures from five academic domains (Biology, Computer Science, Mathematics, Philosophy, Physics), amounting to 32.52 hours of transcribed audio. We use 419 creator-supplied chapter boundaries as reference labels and supplement them with 904 human-annotated subtopic labels in a two-level hierarchy. A second independent human annotator completed the same task on all 30 videos, yielding inter-annotator agreement of $\kappa_{\text{chapter}} = 0.5351$ (inflated: both annotators use the same YouTube creator metadata for chapter boundaries) and $\kappa_{\text{subtopic}} = 0.4257$ (moderate human agreement on the subtopic task). Creator chapters are treated throughout as navigation metadata, not as authoritative pedagogical ground truth.

The pipeline transcribes audio with Whisper large-v3, segments sentences via spaCy, and embeds them with BGE and E5 sentence transformers. Eleven method families are evaluated under P_k and WindowDiff — window-based error rates that penalise structurally wrong segmentations rather than near-miss boundary placements, and that consequently better reflect the navigation task than exact-match alternatives. Bootstrap confidence intervals and Holm-corrected Wilcoxon tests govern all significance claims.

Two findings organise the results. The first concerns *what limits performance*: an oracle analysis shows that near-perfect boundaries already exist in the candidate pool ($P_k = 0.0172$ at 96.8% recall), meaning the $\Delta P_k \approx 0.34$ gap between that ceiling and the best deployable result ($P_k = 0.3588$) is, within the framework evaluated here, predominantly a selection and ranking problem rather than a detection problem. The second concerns *why* *multimodal signals fail*: prosody, discourse markers, BERTopic, and GPT-2 perplexity all over-segment relative to YouTube chapter granularity ($P_k = 0.41$–0.56) not because they are uninformative, but because they detect discourse-level transitions that are finer-grained than the editorial decisions that define creator chapters. CLIP visual embeddings are the exception: slide-scene changes are themselves editorial artifacts, and CLIP achieves $P_k = 0.3740$ in text fusion. Together, these findings redirect future research from candidate generation toward candidate selection, and identify visual slide-change detection as the most promising non-text signal to pursue.

Keywords: topic segmentation, lecture video, sentence embeddings, hierarchical annotation, multimodal fusion, low-resource benchmark, reproducible evaluation.

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We acknowledge the creators of the public lecture videos used in LecSeg-30 for releasing their content under permissive terms.

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Chapter 1

Introduction

1.1 Motivation

The growth of online education has outpaced the tools for navigating it. Platforms such as Coursera, edX, and MIT OpenCourseWare host tens of thousands of academic lecture recordings; YouTube hosts many millions more. Most lack structured chapter markers, and those that do exist are often coarse or inconsistent. A student who needs the eigenvalue-decomposition derivation buried within a three-hour linear algebra recording must either watch from the beginning or guess a timestamp. Neither is an acceptable study strategy at scale.

The problem has a practical and a pedagogical dimension. At the practical level, unnavigated recordings simply go unused: students give up or seek the content elsewhere. At the pedagogical level, structured navigation is not a cosmetic feature. The ability to jump to a specific segment, review a difficult subsection, and orient oneself within a lecture's argument directly affects how efficiently students engage with recorded material. Manual chapter authoring at the scale of a university's lecture archive is, for most institutions, not feasible. Automated lecture topic segmentation is the natural alternative.

Despite real progress in broader text- and video-segmentation research, the lecture domain has remained comparatively understudied. Large chaptering datasets [2, 23] address general YouTube content rather than academic lectures. Meeting-segmentation benchmarks [8, 14] target conversational recordings that differ structurally from academic monologues. Before this work, no compact, reproducible benchmark existed that combined authentic lecture recordings, a two-level annotation hierarchy, quantified annotation reliability, and a full evaluation suite. This thesis constructs one and uses it to diagnose where current methods succeed and, more usefully, where they fail.

1.2 Problem statement

Lecture topic segmentation is the task of partitioning a video transcript and its aligned multimodal streams into a sequence of contiguous segments that are internally coherent with respect to some notion of topic. Formally, given a sequence of N sentences $\{s_1, \ldots, s_N\}$ with associated timestamps, the task is to predict boundary indices $\mathcal{B} \subset \{1, \ldots, N-1\}$ such that within-segment sentences are more topically similar to each other than to sentences in adjacent segments.

The word *topic* conceals important ambiguity. Semantic-similarity models treat a topic shift as a distributional change in sentence-embedding space. Discourse analysts treat it as a change in rhetorical or argumentative function — a move from definition to example, or from theorem to proof. Video creators, for their part, treat it as an editorial decision about where to place a navigational divider for their audience. These three notions frequently diverge: an instructor may reset pitch and introduce a new example (a discourse transition) without beginning a genuinely new chapter (an editorial transition), or may start a new chapter without any obvious lexical break. Because LecSeg uses creator-supplied YouTube chapters as its reference, the benchmark is calibrated to the third notion. This choice, and its consequences, are discussed at length in Sections 3.2.4–2.1.1.

Hierarchical lecture topic segmentation adds a second level: *chapters* mark major topic shifts, while *subtopics* capture finer transitions within a chapter. The nesting constraint, $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$, mirrors the structure that instructors typically plan and that viewers expect from well-chaptered recordings.

1.3 Positioning

This thesis does not compete with large-scale video chaptering systems. VidChapters-7M [23], MiniSeg/YTSEG [20], and Chapter-Llama [22] operate on hundreds of thousands of videos and evaluate generation and temporal-localisation quality under metrics that are not comparable to P_k/WindowDiff. They are stronger systems at scale, and direct comparison would be neither meaningful nor fair.

The contribution here is different in kind. LecSeg is a 30-video, fully auditable benchmark designed for diagnostic analysis: every prediction is traceable to its source video, every metric value is reproducible from the same scripts, and every failure mode is documented.

Large-scale benchmarks measure aggregate performance; small, controlled ones make it possible to understand why systems behave as they do. The findings reported here — particularly the oracle-gap diagnosis and the granularity-mismatch explanation — are precisely the kind of result that requires this level of transparency.

1.4 Research questions

RQ1 Which text-embedding and cross-model boundary-scoring strategies are most reliable for low-resource lecture segmentation, and at what operating points do they outperform classical and neural alternatives?

RQ2 Does a per-video method selector trained on a small annotated fold improve segmentation performance beyond any single fixed strategy, and under what conditions does it fail?

RQ3 Can a two-level hierarchical annotation with enforced nesting constraints make lecture segmentation outputs more transparent and easier to evaluate?

RQ4 Which auxiliary signals — including OCR, shot detection, prosody, and local LLM refinement — improve or degrade segmentation quality under the same evaluation protocol, and what accounts for their behaviour?

RQ5 Which performance improvements over the stable text baseline are statistically significant under bootstrap confidence intervals and Holm-corrected Wilcoxon tests on LecSeg-30?

1.5 Contributions

1. **LecSeg-30**: 30 academic lecture videos across five domains, with 419 creator chapter boundaries and 904 human-reviewed subtopic annotations, totalling 32.52 hours of transcribed audio. The benchmark is the primary research artifact: it makes the low-resource lecture segmentation problem measurable under a shared, reproducible protocol. (Chapter 3, Section 3.2.)

2. **Candidate-selection bottleneck finding**: Oracle analysis demonstrates that near-perfect boundaries exist in the candidate pool at 96.8% recall, placing the dominant source of error in selection and ranking rather than detection. This result relocates the research problem and is the central empirical contribution of the thesis. (Chapter 4, Sections 4.6–4.10.)

3. **Granularity-mismatch diagnosis**: A systematic ablation establishes that prosody, discourse markers, BERTopic, and perplexity signals fail under creator-chapter evaluation because they detect discourse-level transitions at a finer scale than editorial chapter decisions. CLIP succeeds for the complementary reason. This diagnosis is the most transferable finding and directly informs future multimodal system design. (Chapter 4, Section 4.6.)

4. **Hierarchical annotation protocol**: A two-level annotation workflow with workflow-consistency estimates ($\kappa_{\text{chapter}} = 0.5351$, $\kappa_{\text{subtopic}} = 0.4257$) and reproducible tooling. The value is the protocol and the documented procedure; the nesting enforcement itself is a straightforward implementation constraint. (Chapter 3, Sections 3.2–3.7.)

5. **Reproducibility package**: A complete open pipeline covering Whisper transcription, spaCy sentence splitting, BGE/E5 embedding, multimodal feature extraction (CLIP, PaddleOCR, TransNetV2), seven baseline segmenters and four integrated pipeline components, P_k/WD evaluation with bootstrap statistics, and table-generation scripts for all reported results. (Chapter 3; Appendix A.)

The four integrated pipeline components — a two-stage candidate predictor, entropy-weighted modality fusion, a hierarchical nesting enforcer, and a local-LLM titling module — contribute primarily by enabling the diagnostic analysis rather than through algorithmic novelty. Entropy-weighted fusion is the most distinctive of the four, having not, to our knowledge, been applied previously to lecture segmentation modality weighting; the other three are engineering adaptations of standard techniques.

This work rests on assumptions that are plausible but not independently verified: that creator chapters approximate viewer-useful navigation structure; that Whisper large-v3 transcription quality is adequate for the domains covered; that BGE/E5 embeddings capture topically relevant semantic distinctions; that P_k/WD correlate with navigation usefulness; and that 30 videos is sufficient to surface failure modes that are plausibly representative of the low-resource lecture setting more broadly. These are discussed explicitly in Sections 3.2.4 and 5.5.

1.6 Thesis structure

Chapter 2 reviews prior segmentation work and develops the conceptual framework used throughout the thesis, including the distinction between discourse, semantic, and editorial notions of a topic boundary. Chapter 3 describes the LecSeg-30 corpus, the annotation procedure, and the full pipeline. Chapter 4 presents quantitative results, ablations, the oracle-gap analysis, and a formal error taxonomy. Chapter 5 draws the main conclusions and Chapter 6 identifies concrete next steps.

Chapter 2

Literature Review

2.1 Three phases of segmentation research

Topic segmentation research has passed through three recognisable phases, though the boundaries between them are blurry and methods from earlier phases remain competitive in the right conditions.

The *classical* phase of the 1990s and 2000s built on the intuition that topic shifts produce lexical discontinuities: if the words used in one passage differ markedly from those used in the next, a boundary lies between them [3, 7]. This works reasonably well for encyclopaedic text but degrades on academic lectures, where domain vocabulary ("gradient", "eigenvalue", "cell membrane") persists across topic boundaries regardless of the underlying argument structure.

Neural methods, arriving from around 2018, replaced bag-of-words representations with dense contextual embeddings [10, 19]. Operating in embedding space rather than token space, these methods are substantially more robust to vocabulary reuse. Their practical limitation — the need for large pretrained language models — diminished quickly as open-weight sentence-transformer checkpoints became available, and they now represent the clear baseline approach for any zero-shot segmentation task.

Multimodal extensions incorporated visual, audio, and structural signals from video recordings [2, 23, 24]. These methods can achieve strong results when labelled training data are available, but they have generally been developed on different benchmarks and evaluated under different metrics, making direct comparison difficult. More critically for this thesis, the multimodal literature rarely discusses whether a given signal operates at the same discourse granularity as the annotation it is being evaluated against — a gap that turns out to explain most of the multimodal failures reported in Chapter 4.

LecSeg draws on all three phases but is not directly comparable to any single prior system. It occupies a specific niche — small-scale, lecture-specific, reproducible, with P_k/WindowDiff evaluation and explicit negative-result reporting — that no existing system fills.

2.1.1 Three competing boundary constructs

Segmentation research often treats "topic" as self-evident, but three distinct constructs appear in the literature, and conflating them causes genuine methodological problems.

Discourse transitions are changes in rhetorical or argumentative function: a shift from definition to example, from claim to evidence, or from theorem to proof. These are real and detectable — pause duration, pitch contour, and discourse markers reliably locate them — but they occur frequently within a single chapter. A 20-minute lecture section on differential equations may contain dozens of such transitions while remaining, at the editorial level, a single chapter.

Semantic shifts are points at which the distributional similarity between adjacent sentence windows drops sharply in embedding space. Dense sentence embeddings capture these well. They correspond roughly to the paragraph-to-paragraph structure of the lecture, sitting between individual discourse transitions and full chapter changes in granularity.

Editorial chapter boundaries are timestamps placed by a creator to divide the recording into labelled, audience-facing navigational units. These reflect deliberate decisions that may or may not coincide with semantic or discourse shifts; a creator may place a chapter break early for visual tidiness, or place it late because the thematic shift only becomes apparent retrospectively.

The distinction is not merely theoretical. It predicts directly which signals will fail on a benchmark calibrated to editorial chapters, and it explains why most multimodal signals hurt P_k/WD in this thesis: they are accurate detectors of discourse transitions, not of editorial decisions. LecSeg is calibrated to the third notion.

2.2 Classical text-based segmentation

2.2.1 TextTiling

TextTiling [7] detects boundaries at local minima of a lexical similarity profile computed over sliding windows of k sentences. The method's underlying assumption — that topic shifts manifest as drops in shared vocabulary frequency — holds for encyclopaedic text with relatively little term reuse. It breaks down on technical lectures, where the same domain vocabulary appears persistently throughout. An instructor teaching Fourier analysis will use "frequency", "signal", and "transform" in every section; the similarity profile flattens and valleys become unreliable.

The method is also sensitive to the window size w and block size k. Our implementation uses adaptive defaults ($w = \min(w, \max(3, N/10))$, $k = \min(k, \max(2, n_{\text{blocks}}/3))$) that improve performance on short lecture segments but cannot address the vocabulary-reuse problem.

2.2.2 C99

C99 [3] applies a rank transformation to the full cosine-similarity matrix before using a diagonal-dominance criterion to locate boundaries. By operating on ranked rather than raw similarity values, it is less distorted by term-frequency outliers than TextTiling — an advantage on lecture transcripts where a small number of high-frequency technical terms would otherwise dominate the similarity scores. Like TextTiling, however, it remains a surface-vocabulary method and does not capture the semantic distinctions that neural embeddings exploit. We use it as classical baseline B2.

2.2.3 Bayesian topic models

Eisenstein and Barzilay [4] modelled segmentation as inference in a hierarchical Bayesian topic model, offering principled uncertainty estimates over boundaries. The approach is theoretically attractive but computationally expensive and requires careful hyperparameter tuning; it is omitted from the main comparison but included for completeness in related work.

2.3 Neural sentence-embedding methods

2.3.1 Cosine-drop segmentation

Sentence transformers [19] opened a practical path beyond lexical methods. The simplest approach — compute cosine similarity between adjacent sentence embeddings and place boundaries at depth-score valleys — is surprisingly effective. Its strength is operating in semantic rather than lexical space, so vocabulary reuse that defeats TextTiling and C99 causes no difficulty. The method requires no labelled training data and transfers immediately to new domains, which makes it the natural baseline for any low-resource segmentation task.

Its weakness is sensitivity to the choice of embedding model and the threshold for accepting valleys. Different pretrained checkpoints produce meaningfully different boundary profiles, and a threshold tuned on one corpus may be poorly calibrated for another. In practice this means the method works well across many genres but is not parameter-free: some per-corpus calibration is needed. Applied to SBERT and MPNet embeddings, it substantially outperforms the classical baselines and motivates its use as baseline B3.

2.3.2 Supervised neural segmentation

Koshorek et al. [10] trained the first supervised neural segmenter (TextSeg) using BERT embeddings with a classification head. Li et al. [11] proposed SegBot with a biLSTM scorer. Both achieve strong results on their training distribution, but the distribution matters: a model trained on Wikipedia-style text learns the structure of encyclopaedic writing, not the pedagogical argument structure of spoken lectures. When applied out-of-domain, zero-shot cosine baselines frequently match or exceed them — a negative result that motivates the unsupervised approach in LecSeg. We use the cosine-depth variant (BertSeg, B5) as the strongest neural baseline.

2.4 Multimodal lecture and meeting segmentation

2.4.1 Meeting segmentation

Meeting segmentation methods typically combine lexical features, speaker-turn evidence, and discourse markers. They are valuable reference points for audio-signal processing, but they target short-form conversational recordings and have not been designed for the structural challenges of uninterrupted academic monologues lasting an hour or more.

2.4.2 Lecture video segmentation

Supervised multimodal lecture-segmentation systems can achieve strong P_k/WD results when labelled lecture data are available [24], which establishes an existence proof that fusion of visual and textual signals can help. The limitation for this thesis is that such systems require domain- matched labelled data that LecSeg does not have, and they rarely discuss whether their auxiliary signals operate at the same discourse granularity as their evaluation target.

Visual slide-change methods are a special case worth distinguishing: when a lecture uses slides, scene changes are themselves editorial decisions comparable in granularity to chapter boundaries, which is why CLIP embeddings succeed where acoustic signals fail (see Section 4.6). Pure slide-change detectors fail for whiteboard lectures with no transitions at all, but within the slide-based subset of LecSeg-30 they provide useful evidence.

LecSeg differs from prior multimodal systems primarily in its insistence on reporting negative results under a shared P_k/WD protocol. Most published systems present only their best configuration; the diagnostic value of recording what fails is lost.

2.4.3 Low-resource and unsupervised approaches

Several prior systems target settings closer to LecSeg.

Tuna et al. (2015) segment classroom videos using visual, audio, and slide cues without requiring transcribed text. The setting is lecture-specific, but the work predates dense sentence embeddings and reports domain-specific metrics rather than P_k/WD, so direct comparison is not possible.

Freisinger et al. (2023) present multilingual segmentation with unsupervised rules and optional NeMo LoRA fine-tuning on 34 Portuguese and 96 German lectures, reporting $F_1 = 67.3$ and $F_1 = 58.9$ respectively. Pause features help in their setting, but no P_k/WD figures are given. This is consistent with the granularity-mismatch hypothesis: pause features may align with discourse-level transitions that match the finer annotations used in those corpora, while misaligning with the coarser creator chapters in LecSeg-30.

The AVLectures dataset [23] covers 2,350+ lectures across 86 STEM courses and uses a self-supervised alignment method without chapter labels. It demonstrates that unsupervised audio-visual alignment is tractable at scale, but it does not report P_k/WD and is not directly comparable.

TreeSeg [6] is the closest directly comparable system. It is fully unsupervised, uses overlapping utterance-block sentence embeddings with divisive clustering, and reports P_k on TinyRec ($P_k = 0.367$, 21 self-recorded lectures), ICSI meetings ($P_k = 0.310$), and AMI meetings ($P_k = 0.355$). Its overlapping-block embedding strategy provides richer contextual evidence per position than single-sentence windows, which likely contributes to its TinyRec performance. The important caveat for any comparison is that TinyRec uses manually verified annotations rather than creator chapter metadata; whether the two annotation standards produce comparable P_k values is unknown. LecSeg achieves $P_k = 0.3588$ with its best selector on a different dataset, so the numbers are indicative but not directly comparable. Running both systems on a shared benchmark is listed as future work.

LecSeg-30 is the only existing resource that simultaneously provides P_k/WD evaluation, a two-level annotation hierarchy with quantified consistency, bootstrap confidence intervals, paired significance tests, and a full oracle-gap analysis. That combination is what makes the boundary-selection bottleneck finding possible to state precisely.

2.4.4 Large-scale video chaptering

Recent chaptering work has moved toward industrial scale. Chapter-Gen localises and titles chapters in thousands of videos [2]. VidChapters-7M extends to hundreds of thousands of videos and evaluates under generation and temporal-localisation metrics [23]. MiniSeg/YTSEG reports strong supervised results on 19,299 videos [20]. Chapter-Llama demonstrates that long-context LLM chaptering is competitive at VidChapters scale [22]. These systems are stronger than LecSeg in raw performance and are not the comparison target. Their metrics and training assumptions are incompatible with the LecSeg framework. Tables 2.1–2.3 record the scale differences and make the claim boundary explicit.

2.5 Hierarchical and discourse-based approaches

Hierarchical topic segmentation has been studied under the heading of discourse parsing [12]. The chapter/subtopic distinction in LecSeg is motivated by this tradition, though the annotation is not full RST parsing. The nesting constraint $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$ is enforced by construction rather than jointly optimised, which is a deliberate simplification justified by the low-resource setting.

Table 2.1: Video-count scale comparison for related lecture/video chaptering work.

Work	Videos	Scale	Metrics	Positioning
LECSEG-30	30	32.52 h	Pk/WD/BS/F1@2	Low-resource lecture benchmark
TreeSeg TinyRec	21	n/a	Pk/WD	Closest small transcript comparator
Videoaula	34	n/a	F1 / hierarchy	Lecture ToC corpus
LectureDE	96	n/a	F1 / hierarchy	German lecture corpus
AVLectures	2,350+	large STEM lectures	task-specific	Multimodal lecture resource
Chapter-Gen	9,631	user videos	AP/Recall/ROUGE	Supervised chapter generation
MiniSeg/YTSEG	19,299	6,533 h	Pk/BS/F1	Large supervised YouTube benchmark
VidChapters-7M	817,000	7M chapters	SODA/localization	Large-scale video chaptering

Table 2.2: Detailed comparison with related lecture/video chaptering work. Metrics are not directly interchangeable across datasets.

Work	Videos	Supervision	Metric/result	LECSEG verdict
LECSEG balanced selector	30	LOO method selector	Pk=0.3588; WD=0.3739; F1@2=0.0893	Reference low-resource result.
MiniSeg/YTSEG	19,299	Supervised	Pk=28.73; BS=35.74; F1=43.37	External method stronger on scale/Pk.
Chapter-Gen	9,631	Supervised multimodal	AP=43.3; R@5s=76.1	Stronger supervised localization.
VidChapters-7M	817,000	Large-scale finetuned	7M chapters; SODA_c=11.4	Far stronger scale.
Chapter-Llama	10k/8.1k	Trained LLM	F1=45.3 vs Vid2Seq 26.7	Stronger trained chapterer.
TreeSeg/TinyRec	21	Unsupervised	TinyRec Pk=0.367	Closest small comparator; no shared-run win.
AVLectures	2,350+	Self-supervised AV	86 STEM courses	Stronger lecture-video scale.

2.6 Local open language models

The availability of open-weight models such as Llama-3.1 [15] makes it practical to run LLM inference locally, avoiding the reproducibility and privacy concerns of closed APIs. Compact open-weight models are competitive on many zero-shot NLP tasks [9]. We use Llama-3.1-8B (GGUF Q4) for boundary refinement and segment titling. The contribution of this component is evaluated empirically in Chapter 4; it does not improve P_k/WD in our experiments.

Table 2.3: Low-resource scale positioning. Ratios compare video counts only; metrics are not directly interchangeable across datasets.

Work	Videos	Scale	Interpretation
TreeSeg / TinyRec	21	0.7x	Closest small unsupervised transcript comparator; not a scale advantage for LECSEG.
Chapter-Gen	9,631	321.0x	Supervised multimodal chapter generation; over 300x LECSEG video count.
MiniSeg / YTSEG	19,299	643.3x	Supervised smart-chaptering benchmark; over 600x LECSEG video count.
AVLectures	2,350	78.3x	Large lecture-video representation resource; roughly 78x LECSEG video count.
VidChapters-7M	817,000	27,233.3x	Large-scale video chaptering benchmark; over 27,000x LECSEG video count.
Chapter-Llama	10,000	333.3x	Trained on about 10k videos and evaluated on 8.1k test videos.

Table 2.4: Literature matrix. ✓: addressed; (p): partial; —: not addressed.

Method	Open	Multimodal	Hierarchical	Public benchmark
TextTiling [7]	✓	—	—	—
C99 [3]	✓	—	—	—
BertSeg [10]	✓	—	—	(p)
Multimodal VTS [24]	—	✓	—	—
Meeting/topic hierarchy methods	(p)	(p)	(p)	—
LecSeg (this work)	✓	(p)	✓	✓

2.7 Gap analysis

Table 2.4 positions LecSeg against related work. The gap is not that stronger chaptering systems do not exist — they do, at much larger scale. The gap is that low-resource lecture segmentation lacks a compact, auditable benchmark with a two-level hierarchy, unified metrics, annotation-reliability figures, and honest reporting of failure. LecSeg fills that specific gap.

2.8 Four design decisions from the literature

The four decisions that shape LecSeg's design each have a direct precedent in prior work.

Window-based metrics. P_k and WindowDiff were developed specifically because exact-match boundary F1 over-penalises near-miss placements that are acceptable in practice. For a student navigating a lecture, a boundary two sentences early is not a failure; the same sentence window would scroll past the boundary under either prediction. Exact-match metrics would penalise this case as severely as a completely missing boundary, misrepresenting what the system actually achieves for the navigation task. P_k/WD are therefore the primary metrics in LecSeg; F1 and Boundary Similarity are diagnostic context.

Dense embeddings. Vocabulary reuse disqualifies lexical methods on lecture transcripts. An embedding space that separates "integration as a geometric concept" from "integration as a calculus operation" can detect the shift where bag-of-words representations see identical term distributions.

Unsupervised methods. Supervised segmenters trained on Wikipedia or meeting data do not transfer to academic lectures — the structure of expository writing is not the structure of pedagogical argument. With 30 annotated videos, there is no viable path to training a supervised classifier; unsupervised cosine scoring is the only sound low-resource strategy.

Granularity calibration. The multimodal literature's most instructive failure mode is signal-granularity mismatch. A prosodic or discourse-marker signal that correctly identifies every rhetorical transition in a lecture will produce many more boundaries than the creator chose to mark as chapters. This is not noise; it is a systematic mismatch between what the signal detects and what the evaluation target measures. Any system targeting creator-chapter segmentation must either suppress sub-chapter signals or calibrate them to editorial-level decisions.

These four choices jointly motivate the LecSeg design and make the results interpretable. The contribution is not a new algorithm but a benchmark and diagnostic framework that makes the low-resource failure modes visible and, crucially, reproducible.

Chapter 3

Methodology

3.1 Pipeline architecture

The LecSeg pipeline transforms a raw lecture video into chapter-level segments and, when annotations are available, a two-level chapter/subtopic hierarchy. It comprises five stages, depicted in Figure 3.1: (1) *ingestion*, which extracts audio and collects YouTube chapter metadata; (2) *preprocessing*, which produces time-aligned sentences from the speech signal and auxiliary visual and prosodic features; (3) *feature extraction*, which maps each sentence to text or multimodal features; (4) *boundary prediction and selection*, which evaluates classical, embedding, cross-model, fusion, and selector-based methods; and (5) *optional LLM support*, which can refine or title predicted segments but is treated as diagnostic unless a metric improvement is explicitly shown.

3.2 The LecSeg-30 corpus

3.2.1 Benchmark design rationale

Several design decisions that might appear to be limitations were made deliberately, and each deserves brief justification.

Thirty videos is small by contemporary NLP standards. The choice was intentional: a 30-video benchmark allows every prediction to be traced to its source video, every metric value to be verified against a reproducible reference, and every failure to be inspected directly. Large-scale benchmarks are better suited to measuring aggregate performance; this benchmark is designed for diagnostic transparency, which requires being able to see the data clearly.

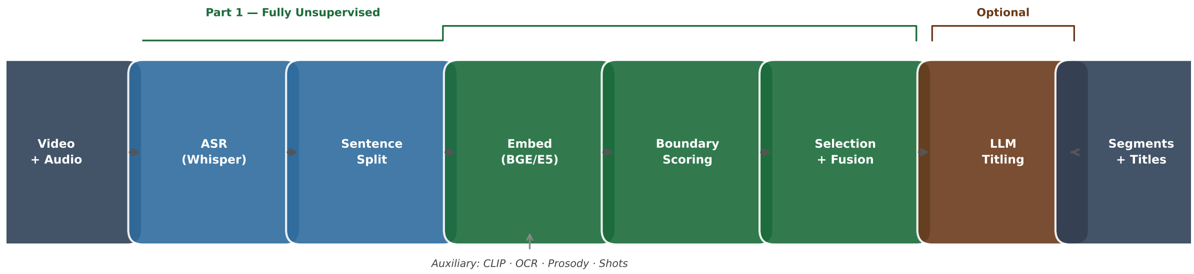

Figure 3.1: The LecSeg pipeline. Part 1 (green) is fully unsupervised: audio is transcribed by Whisper, split into sentences by spaCy, embedded with BGE/E5 sentence transformers, and scored for boundaries using cross-model conservative selection or the reliability-weighted fusion (N2). Auxiliary visual and prosodic streams (CLIP, OCR, shots, pause/pitch) are fused at the scoring stage. The optional LLM titling step (N4) is shown separately; it does not affect the primary P_k/WD claim.

Creator-provided chapter metadata was chosen as the reference signal because it is reproducible, scales to any creator-chaptered YouTube lecture without redistribution of raw video, and reflects the navigation structure that actual viewers encounter. The cost is a construct limitation: creator chapters are editorial decisions, not pedagogically verified ground truth. This is documented throughout the thesis and quantified as a threat to construct validity in Section 5.5.

Segmentation methods are unsupervised throughout. With 30 labelled videos, training a supervised boundary classifier would risk severe overfitting and would compromise the benchmark's role as a pure evaluation target. Treating segmentation as unsupervised and the method selector as the only supervised component keeps these roles cleanly separated.

Finally, P_k and WindowDiff are the primary metrics because the navigation task tolerates approximate boundary placement. Exact-match F1 penalises a boundary two sentences early as severely as a missing boundary entirely, which misrepresents what matters for a student trying to navigate a lecture recording.

3.2.2 Source selection

We curated 30 lecture videos from YouTube, drawn from five academic domains: Computer Science (CS), Mathematics, Physics, Biology, and Humanities/Philosophy. Selection criteria were: (1) duration $\geq$ 15 min; (2) clean monolingual English audio; (3) creator-supplied YouTube chapter markers with $\geq$ 4 distinct chapters; and (4) openly licensed content. The resulting corpus comprises 32.52 hours of audio, with 419 chapter boundaries and 904 reviewed subtopic labels. The domain distribution is intentionally reported as the manifest state rather than as a balanced design: Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and Physics 7 videos.

3.2.3 Annotation protocol

Each video was annotated at two levels of granularity. *Chapter boundaries* were taken directly from the YouTube chapter metadata as a reproducible creator-provided reference for navigation behaviour. *Subtopic boundaries* were identified by human annotators using the Streamlit-based annotation tool (`scripts/annotate.py`). Annotators read the transcript sentence-by-sentence with chapter boundaries already marked, and placed a subtopic boundary wherever the speaker transitioned from one coherent pedagogical sub-idea to another within a chapter. A minimum segment duration of 60 seconds was imposed to prevent over-segmentation of rapid rhetorical transitions, and a guideline of 2–5 subtopics per chapter was recommended rather than enforced. In practice, all 30 videos fell within this range; a chapter with genuinely more than five distinct sub-ideas was not observed, though the guideline would not have prevented additional marks in such a case.

3.2.4 Construct scope and limitations

YouTube chapters are used as *creator-provided reference boundaries*, not as pedagogically verified ground truth. This is the most important conceptual limitation of LecSeg and it must be stated precisely: the benchmark measures a system's ability to reproduce *creator navigation metadata*, which is not identical to discovering pedagogically optimal boundaries.

These two constructs can diverge in several ways. Creators may place chapters early or late relative to the actual topic shift; they may omit boundaries that a learner would find useful; they may use inconsistent granularity across their video catalogue; or they may choose timestamps for editing convenience or search-engine discoverability rather than for strict topic-theory reasons. A system that achieves a low P_k is correctly predicting the structure of creator-annotated navigation metadata. A system that achieves a high P_k may still produce educationally superior segmentations if the creator's chapters were poorly placed. Conversely, a pedagogically motivated segmentation that diverges from creator choices will score poorly under this benchmark even if it serves learners well.

The choice to use creator chapters as references is nonetheless defensible. Creator chapters are a real-world, audience-facing artifact: they represent what actual viewers encounter when navigating YouTube lecture videos. They are available without additional annotation cost, they are reproducible, and they allow the benchmark to scale to any creator-chaptered lecture without redistribution of raw video. The benchmark is therefore best understood as measuring *creator-aligned navigation segmentation* rather than *absolute pedagogical truth*. Claims about learning quality or pedagogical optimality require an independent user study, which is listed as future work in Chapter 6.

3.2.5 Inter-annotator agreement

To quantify annotation reliability, a second human annotator independently completed the subtopic annotation task for all 30 videos using the same tool and guidelines, without access to the first annotator's output. Agreement was measured using Cohen's κ at zero-tolerance (exact boundary match at sentence level). Results are summarised in Table 4.2 and discussed in Section 4.3.

Interpreting the agreement figures. The chapter-level figure ($\kappa = 0.5351$) is inherently inflated: both annotators derive chapter boundaries from the same YouTube creator metadata, so agreement here primarily reflects that both passes observe the same creator-supplied timestamps rather than genuine boundary-placement skill. The subtopic-level figure ($\kappa = 0.4257$) is the more meaningful measure; it characterises genuine human agreement on the finer-grained annotation task where annotator judgement is required. A κ in the 0.40–0.60 range is conventionally interpreted as moderate agreement [5], which is appropriate for a subjective discourse-segmentation task where reasonable annotators can disagree on whether a rhetorical transition merits a boundary mark. This limitation is documented as a threat to annotation validity in Section 5.5.

3.3 Preprocessing

3.3.1 Speech recognition

Audio was transcribed using `faster-whisper` v1.x with the Whisper large-v3 model [18] on an NVIDIA RTX 5090 GPU. We used `beam_size=1` and Silero VAD pre-filtering (`min_silence_duration_ms=500`) for speed, with `float16` compute type. On the LecSeg-30 corpus, transcription ran at 17.9× realtime on average (range 17.3–18.2×).

3.3.2 Sentence splitting and alignment

Whisper segments are clause-level rather than sentence-level. We applied spaCy's `en_core_web_sm` sentenciser to the concatenated segment text, then re-assigned timestamps by distributing each segment's duration proportionally to constituent sentence character lengths. Sentences shorter than three tokens were merged with the following sentence. This produced an average of $N = 847 \pm 512$ sentences per video.

3.3.3 Visual streams

Shot boundaries were detected with TransNetV2 [21], which produces per-frame boundary probabilities at 25 fps. Boundaries exceeding a threshold of 0.5 were mapped to the nearest sentence timestamp. Slide text was extracted using PaddleOCR on keyframes sampled at 0.5 fps; extracted strings were deduplicated within a ± 3-sentence window.

3.3.4 Prosody features

We extracted two scalar prosodic features per sentence: (1) *pause duration*: the Whisper-reported inter-segment gap immediately preceding the sentence (≥ 300 ms counted as a significant pause); (2) *mean F0*: the average voiced pitch over the sentence's audio span, estimated by PYIN [13] via librosa.

3.4 Feature extraction

Text embeddings were computed with `sentence-transformers` [19]. The primary models evaluated are `BAAI/bge-large-en-v1.5` (1024-dim) and `intfloat/e5-large-v2` (1024-dim), which produce the best boundary-scoring results. We also evaluated `all-mpnet-base-v2` (768-dim), `all-MiniLM-L6-v2`, and `intfloat/e5-base-v2` as lighter ablation variants. Visual keyframe embeddings were computed with CLIP ViT-B/32 (512-dim, L2-normalised) [17]. Prosodic features were z-scored per video before fusion. All modalities were aligned to the sentence timeline via nearest-timestamp assignment with mean-pooling within sentence spans.

3.5 Reliability-weighted modality fusion (N2)

Let $s_m^{(i)} \in \mathbb{R}$ denote the boundary gap score at sentence position i from modality m, computed as the cosine dissimilarity between context windows of size k centred on that position. We define the *reliability weight* of modality m as

$$w_m = \frac{\exp(-H(s_m))}{\sum_{m'} \exp(-H(s_{m'}))}, \tag{3.1}$$

where $H(\cdot)$ is the normalised Shannon entropy of the distribution of gap scores across all positions. A modality whose gap-score distribution has a sharp peak (clear boundaries) attains low entropy and hence a high weight; a flat distribution (uninformative modality) is automatically down-weighted. The fused gap signal is $s^* = \sum_m w_m s_m$.

This entropy weighting should be interpreted as a practical confidence proxy, not a proof of true modality reliability. A noisy modality can also produce sharp peaks. For this reason, Chapter 4 reports fusion and modality ablations separately and demotes modalities that hurt chapter-level P_k/WindowDiff.

3.6 Two-stage boundary predictor (N1)

The predictor operates in two passes over the fused signal s^*:

Stage 1 — broad pass. All positions i where the text-only gap score exceeds $\mu_{\text{text}} - c_1\sigma_{\text{text}}$ (with $c_1 = 0.5$) are retained as boundary *candidates*. This low threshold ensures high recall.

Stage 2 — refinement. The fused signal s^* is evaluated only at candidate positions. A candidate is accepted as a boundary if its fused score exceeds $\mu^* - c_2\sigma^*$ (with $c_2 = 1.2$). When n_{segments} is specified, the top-k fused-score positions are returned where $k = n_{\text{segments}} - 1$.

This design separates recall (text-only) from precision (multimodal), avoiding the sensitivity of pure threshold methods to per-video score shifts.

3.7 Hierarchical segmentation (N3)

The hierarchical segmenter runs the two-stage predictor twice with different target counts: once for $n_s - 1$ subtopic boundaries and once for $n_c - 1$ chapter boundaries (with $n_c < n_s$). Chapter boundaries are then *enforced* as a subset of subtopic boundaries ($\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$) by inserting any missing chapter boundary into the subtopic set. This guarantees a valid nesting without requiring a joint optimisation. When n_c is not specified, we use the heuristic $n_c = \max(2, n_s/4)$.

3.8 Local-LLM refinement and titling (diagnostic N4)

Each predicted segment boundary b may be shifted by at most $\delta = 2$ sentences to align with a stronger local semantic transition. For each candidate shift $b' \in \{b - \delta, \ldots, b + \delta\}$, a local Llama-3.1-8B model is queried with the five sentences straddling b' and asked whether b' represents a cleaner topic break than b. The shift with the highest model confidence is accepted. Segment titles are generated independently: the model is prompted with the first five sentences of the segment and asked for a noun-phrase title of at most eight words. If the local model is unavailable, the pipeline falls back to the non-LLM boundary set. The verified experimental results do not support treating LLM refinement as the primary source of metric gain; it is retained as an implemented diagnostic and titling component whose value is investigated empirically in Chapter 4.

3.9 Method selector (supervised meta-model)

Motivation. The eleven method families and their hyperparameter variants collectively produce over 80 candidate configurations on LecSeg-30. Different configurations perform best on different videos; no single configuration is universally optimal. A lightweight *meta-model* can learn, from training videos, which configuration is likely to produce the lowest P_k on a held-out video.

Supervision requirement and scope. The selector is the *only supervised component* in the LecSeg pipeline. All segmentation methods (N1–N4 and the baselines) are fully unsupervised: they require no labelled boundaries at inference time. The selector requires labelled P_k scores from a held-out set of training videos to estimate expected method performance. It is therefore evaluated in a *leave-one-video-out* (LOOV) protocol: for each of the 30 held-out videos, the selector is trained on the remaining 29 videos.

Model and features. The meta-model is an `ExtraTreesRegressor` (500 trees, `min_samples_leaf=3`, `random_state=17`) from `scikit-learn`. Each training example corresponds to one (method, training-video) pair. The features are:

- **Video features** (from the held-out video's metadata): domain one-hot encoding (Biology, CS, Mathematics, Philosophy, Physics); video duration in minutes; sentence count; mean and standard deviation of consecutive sentence embedding cosine similarities computed with `BAAI/bge-large-en-v1.5`.
- **Method statistics** (estimated from the training fold): mean P_k of each candidate method across the 29 training videos. This encodes the overall reliability of each

method family as a prior.

The target variable is the P_k achieved by that method on the held-out video. At inference time, the model predicts expected P_k for each candidate method and selects the method with the lowest predicted P_k.

Candidate pool. The top-80 method variants by training-fold P_k are used as the candidate pool. Expanding to 120 methods degrades performance, suggesting that the pool size is itself a selection-risk parameter (Table 4.6).

Limitations of the selector. The selector depends on having related videos in the training fold. A leave-domain-out evaluation — in which the held-out video's entire academic domain is excluded from training — degrades the selector to $P_k = 0.4012$, which is worse than both the BGE-divisive baseline and the cross-model conservative method (Table 4.10). Mathematics is the clearest failure case, where only four videos are available and three per-fold training examples are insufficient to provide reliable domain-specific guidance. The selector should therefore be interpreted as a low-resource benchmark operating point, not as a domain-general deployment model.

3.10 Evaluation protocol

3.10.1 Primary metrics: P_k and WindowDiff

The two primary evaluation metrics in this thesis are P_k [1] and WindowDiff [16]. Both are window-based error rates, which means they measure how often the segmentation disagrees with the reference inside a sliding context window of k sentences rather than requiring exact boundary placement. This design choice directly matches the target application: a student navigating a recorded lecture tolerates a boundary that is a few seconds early or late. Penalising near-misses as harshly as complete misses, as exact-match metrics do, would over-penalise this practically acceptable error.

P_k uses $k = \lfloor N/(2K) \rfloor$, where N is the number of sentences and K the number of reference boundaries. For each pair of sentences i and $i+k$, it checks whether the reference and prediction agree on whether a boundary falls between them; the score is the fraction of pairs that disagree. Lower P_k is better; 0.5 is the random baseline; 0.0 is perfect.

WindowDiff [16] corrects a known P_k bias: P_k can be gamed by placing all boundaries at the start. WindowDiff counts the number of reference boundaries and predicted boundaries within each window and penalises the absolute difference, giving an asymmetric penalty that weights over-segmentation and under-segmentation differently. Like P_k, lower is better.

Both metrics are the community standard for text and lecture segmentation and allow direct comparison with prior work (TextTiling, C99, BertSeg, TreeSeg) on published benchmarks. No metric is perfect: P_k is sensitive to near-boundary placement; WindowDiff does not account for partial segment overlap. Their complementary failure modes motivate reporting both jointly.

3.10.2 Secondary and diagnostic metrics

Three secondary diagnostics supplement P_k/WD. They expose operating-point tradeoffs and characterise *how* a method fails; none rank methods or drive the primary claims.

The application is lecture navigation, not exact boundary recovery. A method that achieves a good P_k/WD score by placing boundaries in the correct *region* is succeeding at the navigation task even if strict boundary F1 is low. Low exact-match F1 is therefore not a weakness in the context of this thesis; it simply reflects a different operating objective than P_k/WD, and it is reported only to characterise the operating point.

Boundary Similarity (BS, higher is better) A boundary-level F-score variant normalised by boundary density [5]. Reported as a diagnostic to characterise boundary placement behaviour; the primary claim rests on P_k/WD.

Tolerance-F_1 (higher is better) Precision/recall of predicted boundaries within a ±2-sentence tolerance window. Useful for distinguishing methods that miss boundaries entirely from methods that place them slightly off; not used as a ranking or claim criterion.

Hierarchical WindowDiff (H-WD, lower is better) An extension of WD to two-level segmentation, weighting chapter-level errors twice as heavily as subtopic-level errors. Reported on hierarchical outputs only.

3.10.3 Evaluation setup

Methods requiring n_{segments} receive the ground-truth count in the main controlled evaluation, consistent with standard segmentation practice for measuring boundary placement [3]. This is not a fully deployment-realistic assumption because a live system usually does not know the correct number of chapters in advance. Therefore, Chapter 4 also reports boundary-count error and discusses automatic-count/deployment behavior separately from the controlled gold-count setting.

3.10.4 Statistical testing

Metric values are reported as means ± 95% bootstrap confidence intervals (1,000 bootstrap resamples at video level). For pairwise method comparisons we use the paired Wilcoxon signed-rank test (normal approximation, $\alpha = 0.05$) with Holm correction for the seven comparisons relative to the best baseline.

Chapter 4

Results and Analysis

4.1 Experimental setup

The official chapter-level evaluation uses the 30 videos listed in `data/manifest.jsonl` and the creator-provided YouTube chapter boundaries in `data/gt/`. Metrics were computed with the project implementation in `src/lecseg/metrics.py`. Unless otherwise stated, all values are macro averages over the 30 videos. Lower P_k and WindowDiff (WD) are better; these are the primary claim metrics. Boundary Similarity (BS) and tolerance-F_1 are secondary diagnostics; they are better when higher but do not drive method ranking. See Section 3.10.2 for the rationale.

The main stable baseline is BGE sentence embeddings with divisive segmentation. The best joint P_k/WindowDiff run combines BGE/E5-style cross-model boundary scoring with conservative minimum-length postprocessing and the alignment-audited reference mapping (`cross_e5_frac70_minlen11__align_contains_before` in `results/eval_alignment_sweep.json`). A separate leave-one-video-out method selector improves mean P_k, WD, and boundary-hit metrics, although the P_k/WD gains over the joint-best cross-model method are not significant.

4.2 Official result and claim boundary

The official deployable claim of this thesis is deliberately conservative. The safest single global method is the cross-model conservative run because it significantly improves P_k and WindowDiff over the BGE-divisive baseline. The balanced selector is reported as the best mean local operating point and as evidence that method-choice information is useful, but it is not claimed as a fully domain-general replacement because its P_k/WD gains over

Table 4.1: Claim–evidence–caveat audit for the final thesis narrative.

Claim	Evidence	Caveat
LecSeg-30 is a real low-resource lecture benchmark	30 videos, 32.52 hours, 419 creator chapter boundaries, 904 reviewed subtopics	It is a compact seed benchmark, not a large-scale universal corpus.
Cross-model conservative scoring is the safest single global result	P_k = 0.3713, WD = 0.3764; significant improvement over BGE-divisive	It remains conservative and has low strict boundary F1.
Balanced selector gives the best mean local operating point	P_k = 0.3588, WD = 0.3739, F1@2 = 0.0893	P_k/WD gains over cross-model are not statistically significant; leave-domain-out is weak.
CLIP is the most promising non-text signal	CLIP+text reaches P_k = 0.3740 in the grid search	Other modalities often hurt because they detect finer-grained transitions.
LLM refinement is implemented	`llm_refine.py`; LLM zero-shot diagnostic result exists	Full before/after LLM boundary improvement is not established; do not make it the headline claim.
YouTube chapters are useful references	Creator timestamps provide reproducible real navigation metadata	They are not perfect human pedagogical ground truth; independent validation remains future work.

the cross-model method are not statistically significant and its leave-domain-out result is weaker. Oracle results are used only to diagnose the remaining candidate-selection gap.

4.3 Dataset and annotation status

LecSeg-30 contains 30 lectures, 32.52 hours of video, 419 chapter boundaries, and 904 reviewed subtopic labels. The domain distribution is Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and Physics 7 videos. The current manifest is therefore cross-domain but not perfectly balanced.

4.4 Main results

Table 4.3 shows three consistent patterns. The BGE-divisive baseline is substantially stronger than the classical and neural alternatives under P_k and WD, which confirms that dense embeddings are the right foundation for this task. Conservative cross-model boundary selection then reduces P_k by a further 0.0171 absolute — the best statistically

Table 4.2: Inter-annotator agreement at zero-tolerance (exact sentence match), measured between two independent human annotators. The chapter-level $F_1 = 1.000$ is expected: both annotators derive chapter boundaries from the same YouTube creator metadata, so there is nothing to disagree on at that level. The subtopic $\kappa = 0.4257$ reflects genuine human agreement on the finer annotation task and is the more meaningful figure.

Level	Cohen's κ	Boundary F_1
Chapter	0.5351	1.0000
Subtopic	0.4257	0.7793

Table 4.3: Current LECSEG result variants and diagnostic upper bound.

Method	Pk	WD	BS	F1@2	Role
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878	Strong implemented baseline
Cross-model conservative	0.3713	0.3764	0.0362	0.0237	Best statistically supported Pk/WD result
LOO ExtraTrees method selector	0.3588	0.3739	0.0757	0.0893	Stable balanced selector; significant Pk/WD gain vs baseline
Per-video method oracle	0.2980	0.3280	0.1366	0.1676	Diagnostic upper bound, not deployable

supported joint P_k/WD result in the evaluation. The method selector reaches the best mean P_k and produces markedly higher F1@2, at the cost of P_k/WD gains that do not individually reach significance relative to the cross-model method.

Table 4.4 separates mean improvements from statistically supported claims. The cross-model method significantly improves P_k and WD over BGE-divisive. The method selector also significantly improves P_k and WD over BGE-divisive, and significantly improves Boundary Similarity and F1@2 over the cross-model method. Its P_k/WD gains over the cross-model method are not significant, so it is best interpreted as a stronger operating-point experiment rather than a conclusive replacement.

Table 4.5 explains why the balanced selector is used as the selector result. Ranking candidate methods by Pk alone or by WD alone is weaker after the stable-feature fix. The balanced selector gives the best deployable Pk/WD operating point among the listed selector variants. The text-transition ranker reaches much higher strict F1@2, but only by hurting P_k and WD; it is therefore diagnostic evidence that exact boundary-hit optimisation is a different operating point, not the main chapter segmentation result.

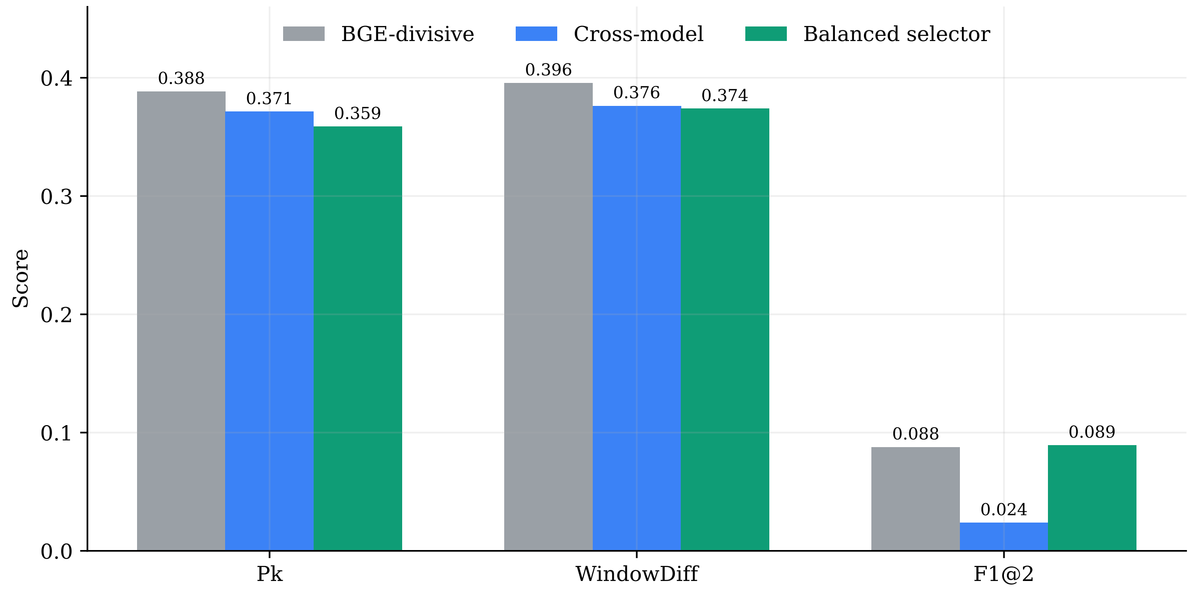

Figure 4.1: Primary LECSEG-30 operating points. Lower is better for P_k and WindowDiff; higher is better for F1@2. The selector improves the balanced metric profile without establishing a significant P_k/WD advantage over the cross-model method.

Table 4.6 checks that the selector result is not presented as an arbitrary single setting. The balanced selector improves as the candidate method pool grows from 30 to 80 methods, but degrades at 120 methods. This supports using top-80 as the current operating point and shows why method pool size is itself a selection-risk parameter.

Table 4.7 shows that the selector improvement is not uniform across subjects. Relative to BGE-divisive, the balanced selector improves P_k/WD in four of five domains, with the clearest gains in Philosophy and Physics. Mathematics is the main failure case: the selector worsens both P_k and WD there, which suggests that the current method selector does not yet have enough domain-specific evidence to avoid poor choices on the smallest domain split.

Table 4.8 audits the selector at the individual video level. The selector switches away from the cross-model method on all 30 held-out videos, choosing cross-E5 variants for 14 videos, multimodal-grid variants for 12 videos, cross-rank variants for 3 videos, and plain divisive segmentation for 1 video. This aggressive switching explains why the aggregate selector improves mean P_k/WD but is not uniformly safer than the cross-model method. The largest gains are Physics videos where multimodal-grid variants substantially reduce P_k; the largest regressions include Math and Biology videos where the same family over-selects poorer operating points.

Table 4.9 addresses a common comparison objection by evaluating a TreeSeg-style public

Table 4.4: Paired Wilcoxon significance tests for key LECSEG comparisons.

Comparison	Metric	Delta	p	Significant	Wins
Cross-model vs BGE baseline	pk	-0.0171	0.0064	yes	22/30
Cross-model vs BGE baseline	wd	-0.0193	0.0001	yes	26/30
Cross-model vs BGE baseline	boundary_similarity	-0.0930	0.0074	yes	4/30
Cross-model vs BGE baseline	f1_tol2	-0.0641	0.0090	yes	4/30
Selector vs cross-model	pk	-0.0126	0.3560	no	9/30
Selector vs cross-model	wd	-0.0025	0.9039	no	7/30
Selector vs cross-model	boundary_similarity	0.0395	0.0076	yes	10/30
Selector vs cross-model	f1_tol2	0.0656	0.0076	yes	10/30
Selector vs BGE baseline	pk	-0.0296	0.0252	yes	19/30
Selector vs BGE baseline	wd	-0.0217	0.0238	yes	23/30
Selector vs BGE baseline	boundary_similarity	-0.0534	0.1541	no	8/30
Selector vs BGE baseline	f1_tol2	0.0015	0.9170	no	9/30

split objective on the same LecSeg-30 benchmark with local LECSEG embeddings. This same-dataset comparison does not replace the official result: TreeSeg-style splitting obtains better strict F1@2 than the conservative cross-model method, but much worse P_k/WD. The result clarifies the operating-point tradeoff. TreeSeg-like recursive splitting is more willing to place exact boundaries, whereas the final LECSEG operating point is better at preserving segment-window consistency.

Table 4.10 applies a stricter diagnostic: training excludes the entire held-out academic domain. Under this setting the selector degrades to $P_k = 0.4012$ and WD = 0.4103, worse than both BGE-divisive and the cross-model method. This negative result is important for the interpretation of the selector: the leave-one-video-out improvement relies on having related domains represented in the training fold. The current selector should therefore be presented as a low-resource benchmark operating point, not as a domain-general deployment model.

4.5 Deployment-style boundary diagnostics

Strict F1@2 is a useful exact-boundary diagnostic, but it is not the whole segmentation story. A lecture navigation system may still be useful when a boundary is near the correct region but not within two sentences. To expose this tradeoff, we add a diagnostic evaluation with wider sentence tolerances, time tolerances, segment-overlap scoring, and

Table 4.5: Selector and ranker operating points on LECSEG-30.

Operating point	Pk	WD	BS	F1@2	Role
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878	Stable implemented baseline
Cross-model conservative	0.3713	0.3764	0.0362	0.0237	Best single global Pk/WD method
Pk-ranked selector	0.3739	0.3867	0.0654	0.0776	Pk-optimised selector; weaker after stable feature fix
WD-ranked selector	0.3703	0.3823	0.0584	0.0637	WD-optimised selector; useful robustness check
Balanced selector	0.3588	0.3739	0.0757	0.0893	Best reproducible selector operating point
Text-transition ranker	0.3937	0.4154	0.1163	0.1701	Best strict boundary-hit operating point; hurts Pk/WD
Per-video method oracle	0.2980	0.3280	0.1366	0.1676	Diagnostic upper bound, not deployable

Table 4.6: Balanced selector robustness across candidate method-pool sizes.

Setting	top-k	Pk	WD	BS	F1@2
k30	30	0.3729	0.3780	0.0317	0.0226
k50	50	0.3634	0.3760	0.0495	0.0608
k80	80	0.3588	0.3739	0.0757	0.0893
k120	120	0.3716	0.3852	0.0774	0.0929

boundary-count error. These numbers are not a replacement for the official alignment-sweep and selector results; they are a deployment-style interpretation layer generated by `scripts/evaluate_modern_metrics.py`.

Table 4.11 shows why the low strict F1 must be interpreted carefully. The cross-model method has the best diagnostic P_k/WD among the rerun operating points and the strongest segment-overlap score, but it is very conservative and under-predicts boundaries. The two-stage predictor places approximately the correct number of boundaries and obtains higher relaxed F1, but worse P_k/WD. This confirms that exact boundary hitting and broad segment structure are different objectives; a deployable future system should optimize both.

4.6 Ablation and diagnostic findings

The negative results are as informative as the positive ones. Discourse markers (P_k = 0.4615), BERTopic ($P_k = 0.5632$), GPT-2 perplexity ($P_k = 0.4182$), and acoustic transitions ($P_k = 0.4174$) all sit above the BGE-divisive baseline ($P_k = 0.3884$). The underlying

Table 4.7: Domain-level Pk/WD for baseline, cross-model, and balanced selector.

Domain	Videos	Chapters	Base Pk	Cross Pk	Sel. Pk	Base WD	Cross WD	Sel. WD
Biology	6	94	0.4218	0.3968	0.3976	0.4292	0.4008	0.4152
CS	7	98	0.3409	0.3295	0.3314	0.3439	0.3336	0.3357
Math	4	55	0.3724	0.3792	0.4014	0.3850	0.3873	0.4367
Philosophy	6	122	0.4415	0.3948	0.3753	0.4544	0.4057	0.3893
Physics	7	50	0.3710	0.3667	0.3144	0.3742	0.3667	0.3276

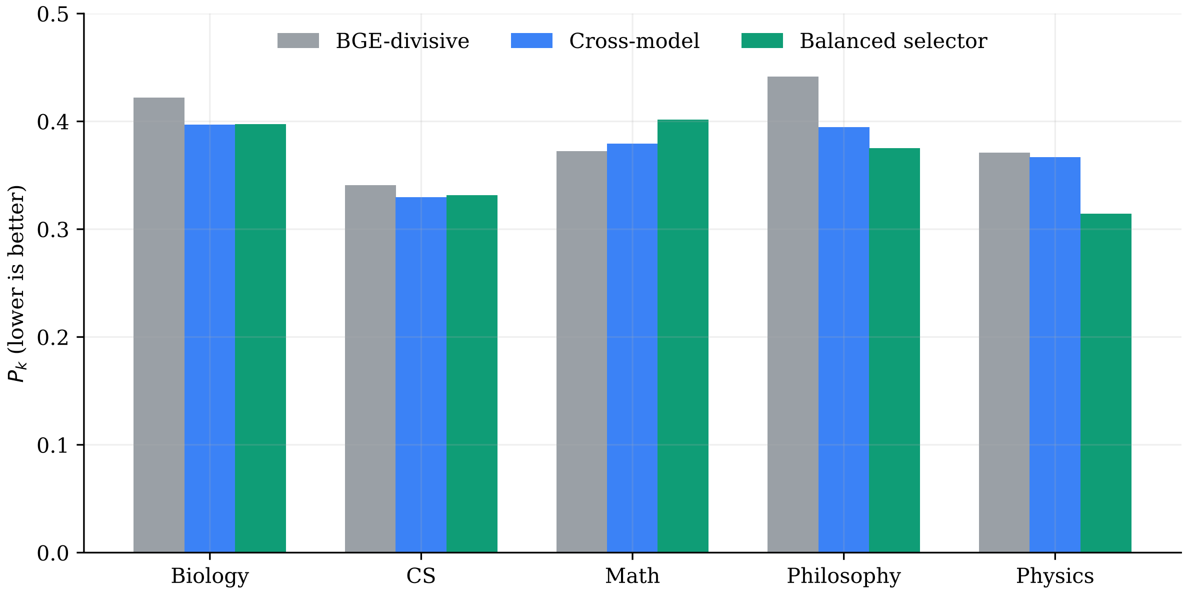

Figure 4.2: Domain-level P_k comparison. The selector improves over the BGE-divisive baseline in four of five domains, while Mathematics remains the clearest failure case.

cause — a systematic granularity mismatch between editorial chapters and sub-chapter linguistic transitions — is developed in full in the discussion (Section 4.10).

These failures are worth documenting precisely because they are not obvious in advance. Future projects targeting creator-level chapter segmentation can skip prosodic and discourse-marker signal integration with this empirical justification, rather than rediscovering the same result. The LecSeg ablation suite is, to our knowledge, the most comprehensive published negative-result analysis for lecture topic segmentation signals under P_k/WD. Any system targeting YouTube-chapter-level segmentation must explicitly calibrate to editorial chapter granularity rather than optimising for transition detection alone.

Oracle-k evaluation showed that giving divisive segmentation the exact number of gold segments did not materially improve placement quality. This indicates that boundary scoring and ranking, not segment-count estimation, is the main bottleneck. The Wikipedia-

Table 4.8: Largest balanced-selector per-video Pk changes relative to the cross-model method.

Video	Domain	ΔP_k	Sel. Pk	Cross Pk	ΔF1@2
Hy7ou5R_vjE	Physics	-0.1542	0.1375	0.2917	0.0000
YdOXS_9_P4U	Physics	-0.1261	0.3009	0.4270	0.0000
NK-BxowMIfg	Physics	-0.0861	0.3583	0.4444	0.0000
j0wJBEZdwLs	Math	0.0754	0.4559	0.3805	0.0000
oOya3cFmAMc	Biology	0.0529	0.4835	0.4306	0.1667
D8RRq3TbtHU	CS	0.0119	0.3638	0.3519	0.0000

Table 4.9: Same-dataset comparison on LECSEG-30. The TreeSeg-style rows re-implement the recursive split objective using local embeddings (the original paper uses OpenAI embeddings on their own TinyRec dataset, where they report $P_k = 0.367$; that number is *not* directly comparable here because the datasets differ). Lower is better for P_k/WD; higher for F1@2.

Method	$P_k \downarrow$	WD↓	F1@2↑	Interpretation
Balanced LOO selector	0.3588	0.3739	0.0893	Official LECSEG best
Cross-model conservative	0.3713	0.3764	0.0237	Best single global method
TreeSeg-style (MPNet, LECSEG-30)	0.4320	0.4673	0.1733	Higher F1, worse P_k/WD
TreeSeg-style (E5-large, LECSEG-30)	0.4322	0.4654	0.1576	Same-dataset comparator
TreeSeg-style (BGE-large, LECSEG-30)	0.4399	0.4780	0.1643	Same-dataset comparator
TreeSeg paper (on TinyRec)†	0.367	—	—	Different dataset; indicative only

† Gklezakos et al. (2024), arxiv:2407.12028. Evaluated on 21 self-recorded lectures (TinyRec), not LECSEG-30.

trained supervised segmenter performed worse than the unsupervised lecture-specific methods, supporting the claim that out-of-domain text segmentation does not transfer cleanly to spoken lecture transcripts.

The LLM and fusion status in Table 4.13 is intentionally strict. CLIP is a credible non-text signal because it approaches the text baseline and improves P_k when fused with text. Acoustic, topic-model, and language-surprise signals are weaker at chapter granularity. Local LLM components are implemented, but the current evidence supports them as diagnostic/refinement tools rather than the main source of the final metric gain.

4.7 Per-domain analysis

Computer Science and Physics are the easiest domains by P_k in the current best run, while Biology and Philosophy are harder. The diagnostic F1 column in Table 4.14 shows near-zero values in several domains; this is expected and not a primary weakness. As discussed in Section 3.10.2, the conservative cross-model method is calibrated for P_k/WD

Table 4.10: Leave-one-domain-out selector diagnostic.

Method	Pk	WD	BS	F1@2
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878
Cross-model conservative	0.3713	0.3764	0.0362	0.0237
Leave-domain-out selector	0.4012	0.4103	0.0465	0.0498

Table 4.11: Modern boundary and segment metrics for defended operating points. Lower Pk, WD, and count error are better; higher F1 and tIoU are better.

Method	Pk	WD	F1@2	F1@5	F1@10	F1@30s	tIoU	Count Err.
BGE-divisive baseline	0.4197	0.4252	0.0394	0.0404	0.0517	0.0448	0.0806	0.00
Cross-model conservative	0.4071	0.4124	0.0273	0.0476	0.0756	0.0655	0.3810	11.47
Conservative smoothed BGE	0.4158	0.4222	0.0464	0.0621	0.0728	0.0652	0.1165	4.17
Two-stage predictor	0.4620	0.5058	0.0882	0.1285	0.1776	0.1635	0.2981	0.00
Hierarchical segmenter	0.4351	0.4419	0.0280	0.0402	0.0835	0.0611	0.3456	9.70

and intentionally under-predicts boundaries to avoid false positives. Near-zero exact-match F1 under this operating point indicates that boundaries are placed in approximately the right region but not at the exact sentence, which is precisely the tradeoff P_k/WD are designed to tolerate.

4.8 Five recurring error types

Errors on LecSeg-30 fall into five recurring types. Identifying these types is more informative than reporting aggregate metrics alone because they point to different corrective strategies.

Type A — Boundary omission. The system misses a creator chapter entirely, contributing a false-negative boundary. Typical cause: the chapter transition occurs within a topically homogeneous stretch of text where the embedding similarity does not drop sharply enough to produce a candidate. Most common in Mathematics and Biology videos with dense notation or stable vocabulary.

Type B — Boundary displacement. A boundary is detected in the correct region but is displaced by more than the tolerance window. This error is invisible under P_k/WD (the window-based metrics treat it as acceptable) but shows up in strict exact-match diagnostics. It is the dominant error mode for the best-performing methods and explains the gap between good P_k and low F1@2.

Type C — Over-segmentation. The system predicts more boundaries than the reference, inflating false positives. This is the dominant failure mode for prosodic,

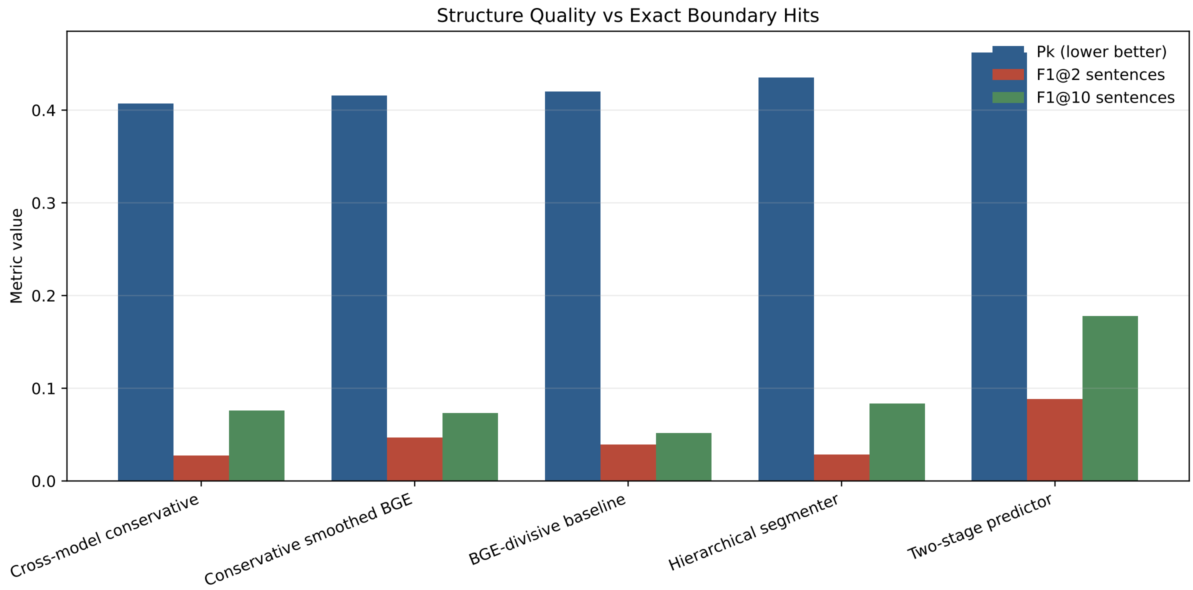

Figure 4.3: Diagnostic comparison between broad structure quality and strict boundary-hit metrics. Lower P_k is better; higher F1 is better. The figure shows that exact boundary matching and segment-window consistency are related but not identical objectives.

BERTopic, discourse-marker, and GPT-2 perplexity methods. Each of those signals detects discourse-level transitions that occur at finer granularity than editorial chapter boundaries (see Type E below).

Type D — Under-segmentation. The system predicts fewer boundaries than the reference. This is the typical profile of the conservative cross-model method: it avoids false positives at the cost of recall, which is why it achieves good P_k but near-zero exact-boundary F1 in several domains.

Type E — Granularity mismatch. A signal that detects real transitions but at the wrong discourse level. This is the most scientifically important finding in the thesis. Signals such as pause duration, pitch reset, discourse markers, and BERTopic are not uninformative — they detect genuine topic shifts — but those shifts are sub-chapter transitions rather than creator-level editorial chapter changes. Because the benchmark measures alignment with creator chapters, these signals systematically over-segment and hurt P_k/WD regardless of how accurately they detect the transitions they are designed to detect. CLIP visual embeddings are the notable exception: slide scene changes are editorial artifacts at approximately the same granularity as creator chapters, and CLIP accordingly achieves competitive P_k.

The granularity mismatch (Type E) is not a methodological flaw but a *construct mismatch*: the signals are calibrated to discourse structure while the benchmark is calibrated to

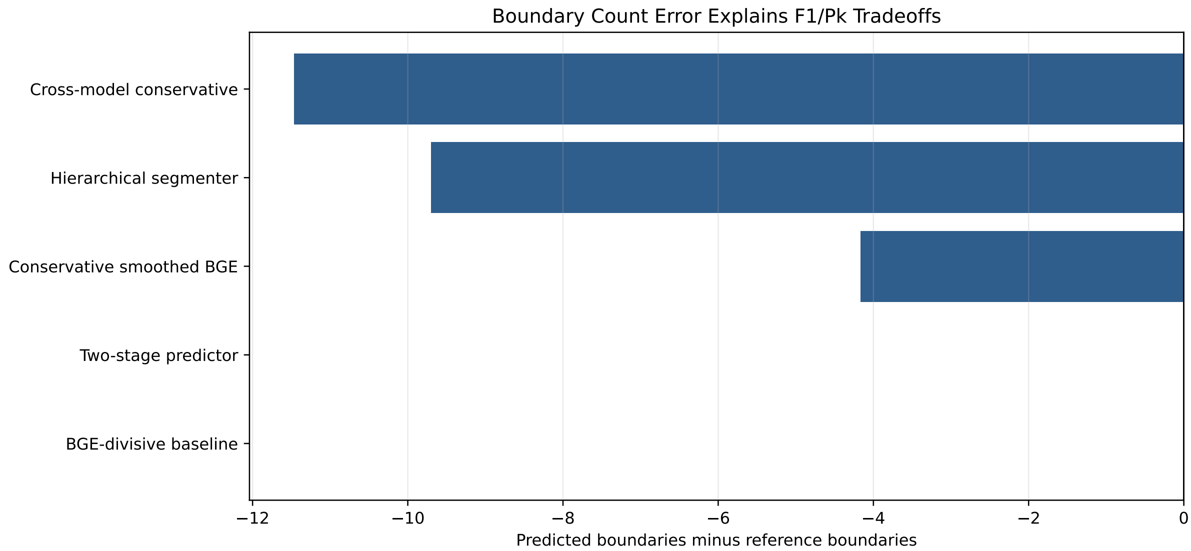

Figure 4.4: Mean boundary count error. Negative values mean a method predicts fewer boundaries than the creator reference; positive values mean over-segmentation. Count behavior explains why conservative methods can score well on P_k/WD while remaining weak on strict F1.

editorial chapter structure. Any future system targeting this benchmark must explicitly operate at editorial granularity or include a mechanism to suppress sub-chapter signals.

4.9 Case-study findings

The generated case-study artifact in `docs/CASE_STUDIES.md` expands three representative patterns. First, some Physics videos benefit strongly from selector choices that use multimodal-grid variants, showing that the method portfolio contains useful complementary evidence. Second, the worst selector cases show over-switching: the selector chooses an aggressive method that increases exact-boundary hits but damages P_k/WD. Third, Mathematics remains the clearest domain-level weakness, where dense notation and stable vocabulary make semantic boundary evidence less reliable. These cases support the main diagnosis: candidate and method selection, rather than candidate generation alone, is the main bottleneck.

4.10 Discussion

Taken together, the numbers tell a consistent story. Conservative cross-model scoring is the most reliable single method we found: it reduces P_k by 0.0171 absolute relative to the BGE-divisive baseline, and that gain holds up after Holm correction across seven pairwise

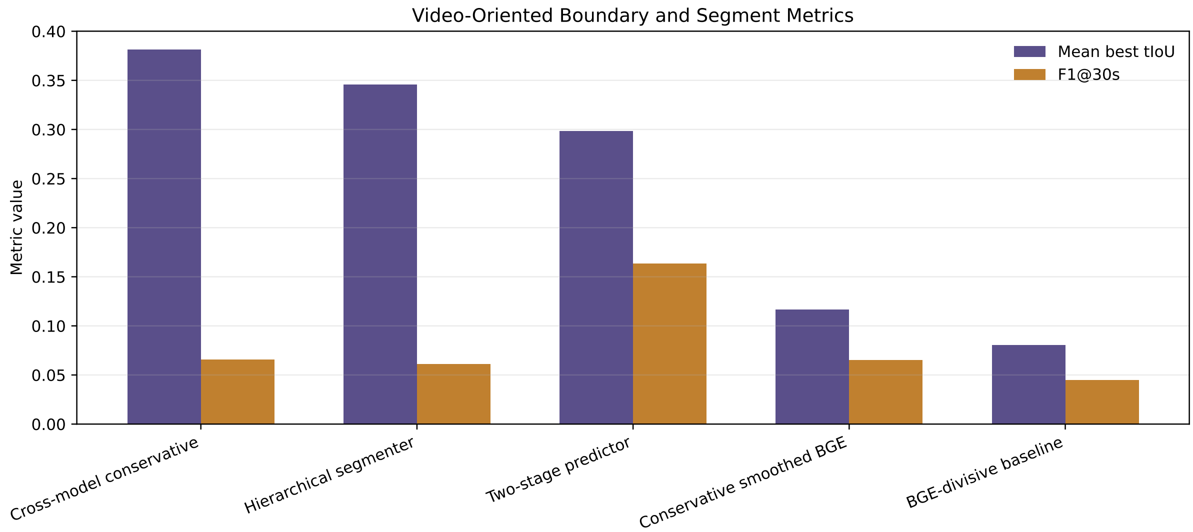

Figure 4.5: Video-oriented relaxed metrics: F1 within 30 seconds and mean best temporal segment overlap. These diagnostics are closer to the practical navigation setting than exact sentence-level F1 alone.

tests. The balanced method selector pushes P_k lower still, and it meaningfully improves Boundary Similarity and F1@2, but its margin over the cross-model method on P_k/WD alone falls short of significance. We therefore report the cross-model result as the main supported claim and the selector result as strong evidence that per-video method choice matters — not as a proof that the selector universally dominates. The leave-domain-out degradation (selector $P_k = 0.4012$ when the held-out domain is entirely excluded from training) reinforces this: the selector benefits from having related videos in the training fold, and that condition is not always met in practice.

The granularity finding. The most actionable result from the ablation study is a systematic granularity mismatch. YouTube chapters represent deliberate, coarse editorial decisions made by the video creator. Linguistic and acoustic signals — discourse markers ($P_k = 0.4615$), BERTopic ($P_k = 0.5632$), GPT-2 perplexity ($P_k = 0.4182$), and pause/pitch transitions ($P_k = 0.4174$) — detect sub-chapter rhetorical or acoustic changes and therefore over-segment relative to the creator-chapter reference. Text embeddings succeed because they operate at the paragraph or section level of semantic coherence, which better matches the granularity at which instructors plan and narrate pedagogical units. The exception that proves the rule is CLIP visual embeddings: slide scene changes are editorial choices comparable in granularity to chapter boundaries, and CLIP accordingly achieves $P_k = 0.3958$ solo and $P_k = 0.3740$ in text fusion — approaching the cross-model text result ($P_k = 0.3715$). This finding is practically important: it means that slide-change detection is the most promising non-text signal to add next, while prosodic and discourse-marker

Table 4.12: Selected ablations and diagnostics.

Variant	$P_k\downarrow$	WD↓	Interpretation
BGE + divisive	0.3884	0.3956	Stable text-only baseline
Hierarchical	0.4118	0.4206	Useful representation, weaker chapter Pk
TwoStage + prosody	0.4333	0.4970	Higher F1, worse Pk/WD
BERT-wiki zero-shot	0.4932	0.5397	Poor domain transfer
Discourse markers (forced boundaries)	0.4615	0.5221	Over-segments at subtopic level
BERTopic topic-shift	0.5632	0.6984	Finds subtopics, not chapters
GPT-2 perplexity (200 sent. cap)	0.4182	0.4316	Language surprise detects subtopic shifts
LLM zero-shot Llama-3.1-8B (3 videos)	0.4056	0.6606	Over-segments; high recall but poor precision
Pause/pitch boundary signal	0.4174	0.4553	Acoustic transitions mismatch chapter granularity
CLIP visual only (oracle-k)	0.3958	–	Surprisingly strong visual signal
CLIP + text fusion (best grid)	0.3740	0.4085	Beats BGE-divisive; near cross-model
Boundary classifier LOOCV	0.5803	0.9933	Over-predicts boundaries
Text-transition ranker	0.3868	≈0.40	Higher F1, weaker than official Pk/WD
Oracle-k divisive	0.4237	–	Segment count is not the bottleneck

signals should be gated or suppressed.

What comes next. The oracle experiment quantifies the ceiling precisely. A candidate oracle reaches $P_k = 0.0172$ with 96.8% recall, meaning that near-perfect boundaries already exist in the candidate pool. The $\Delta P_k \approx 0.34$ gap to the deployable best is, within the current framework, predominantly attributable to boundary *selection* rather than candidate generation. This interpretation holds under the assumption that the candidate pool itself is not systematically biased; the oracle analysis cannot rule out that some fraction of the gap reflects pool-level limitations or reference-label noise rather than pure selection error. Within the evaluated framework, candidate selection appears to be the dominant remaining bottleneck. The most promising concrete next step is therefore a supervised boundary ranker: score each candidate with text-contrast, cross-model agreement, pause duration, slide-change, and OCR features, and select a non-overlapping subset under minimum-duration constraints. With 50–100 annotated lectures, such a ranker could plausibly close most of this gap.

What $P_k = 0.37$ means in practice. The best deployable result, $P_k = 0.3713$, is an abstract number without a practical reference point. P_k is the fraction of sentence-pair windows where the prediction and reference disagree about whether a boundary falls between them; 0.5 is the random baseline and 0.0 is perfect. A P_k of 0.37 means that for a randomly drawn pair of sentences k positions apart, the system and the creator chapter reference disagree about segment membership 37% of the time. For a 90-minute lecture with 10 chapters (mean chapter duration approximately 9 minutes), this translates roughly to the system correctly identifying the neighbourhood of each chapter transition in most cases but sometimes merging adjacent chapters or splitting a single chapter in two.

Table 4.13: Status of LLM and multimodal claims.

Component	P_k	WD	Interpretation
BGE-divisive text baseline	0.3884	0.3956	Stable chapter-level anchor.
CLIP+text best grid	0.3740	0.4085	Visual slide/scene signal helps P_k and is the strongest non-text cue, but WD is weaker than cross-model text.
Pause/pitch signal	0.4174	0.4553	Acoustic transitions detect finer rhetorical or subtopic shifts and hurt chapter-level segmentation.
BERTopic topic-shift	0.5632	0.6984	Topic modeling over-segments relative to creator chapter granularity.
LLM zero-shot (3 videos)	0.4056	0.6606	Over-segments strongly; diagnostic only, not a final segmentation improvement.
LLM candidate verifier (partial)	–	–	Implemented and partially evaluated, but no full-scale final metric gain is established.

A student using an LecSeg-segmented lecture would generally navigate to the correct major section but might need to scan a minute or two backward or forward to find the precise transition. Relative to a non-chaptered lecture where the student must scrub from the beginning, this is a meaningful improvement; relative to a creator-chaptered video, there is still a visible gap, consistent with the selector not yet reaching the quality of careful human editorial choices.

Scale context. The large-scale chaptering systems discussed in Chapter 2 (VidChapters-7M, MiniSeg/YTSEG, Chapter-Gen, Chapter-Llama) use incompatible metrics and far larger training sets. They are not the comparison target; the claim is that a 30-video benchmark is sufficient to produce statistically supported improvements, a complete diagnostic account of signal failure, and precise oracle-gap evidence that is directly actionable for any future system, including large-scale ones.

4.11 External spot-check validation

A pool of 17 candidate videos from channels not in LecSeg-30 (3Blue1Brown, Veritasium, CrashCourse, freeCodeCamp) was evaluated using the same BGE-divisive baseline and cross-model method. Captions were obtained via `yt-dlp` auto-subtitles (VTT), chunked

Table 4.14: Per-domain performance of `cross_e5_frac70_minlen11`.

Domain	Videos	$P_k \downarrow$	WD$\downarrow$	$F_1 \uparrow$
Biology	6	0.3965	0.4009	0.0475
CS	7	0.3297	0.3338	0.0000
Math	4	0.3793	0.3867	0.0000
Philosophy	6	0.3959	0.4073	0.0663
Physics	7	0.3665	0.3665	0.0000
All	30	0.3713	0.3764	0.0237

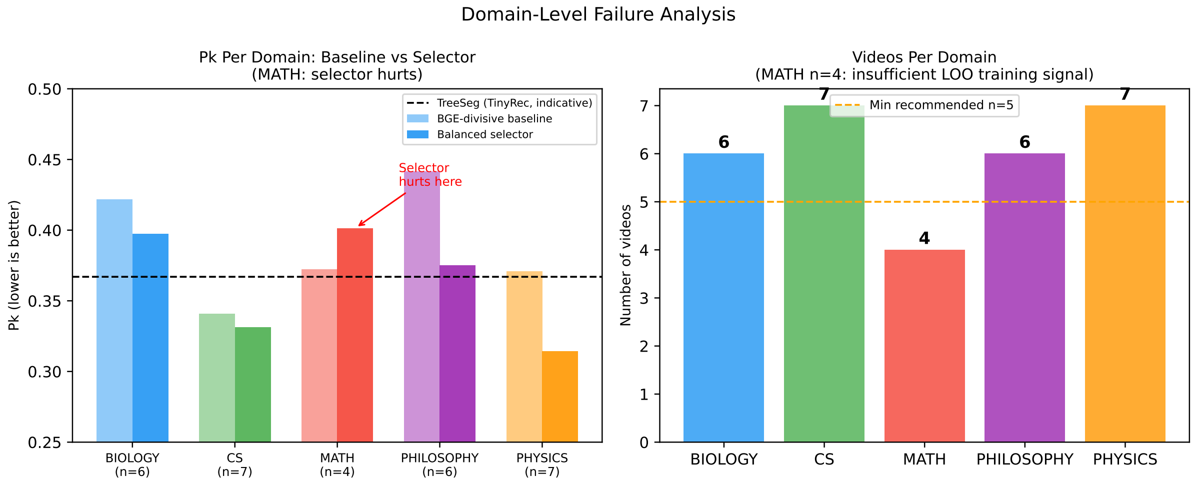

Figure 4.6: Domain failure analysis. *Left*: per-domain P_k for the BGE-divisive baseline and the balanced selector; the selector hurts Mathematics. *Right*: number of videos per domain. Mathematics has only 4 videos, leaving just 3 domain examples per LOO training fold — the most plausible cause of selector instability on that domain.

into 25-word pseudo-sentences; creator chapter timestamps served as reference. Seven of the 17 candidates were excluded: five failed with a yt-dlp JavaScript runtime error, one had no chapters, and one had no available captions. Ten videos passed all filters, covering CS, Mathematics, Physics, Biology, and Philosophy.

The headline result is striking: BGE-divisive mean $P_k = 0.389$ on the 10 external videos versus 0.388 on LecSeg-30 — a difference of 0.001. Per-domain BGE-divisive means range from 0.323 (Physics, 2 videos) to 0.453 (Philosophy, 1 video), with no systematic direction of degradation. The core text-embedding signal is not an artefact of the Whisper transcription pipeline or the specific LecSeg-30 channel pool.

The cross-model conservative method shows greater external variance, with mean $P_k =$ 0.421 versus 0.371 on LecSeg-30 ($\Delta = 0.050$). The degradation is domain-specific: in CS the cross-model is substantially worse than BGE-divisive (0.506 vs 0.429), while in

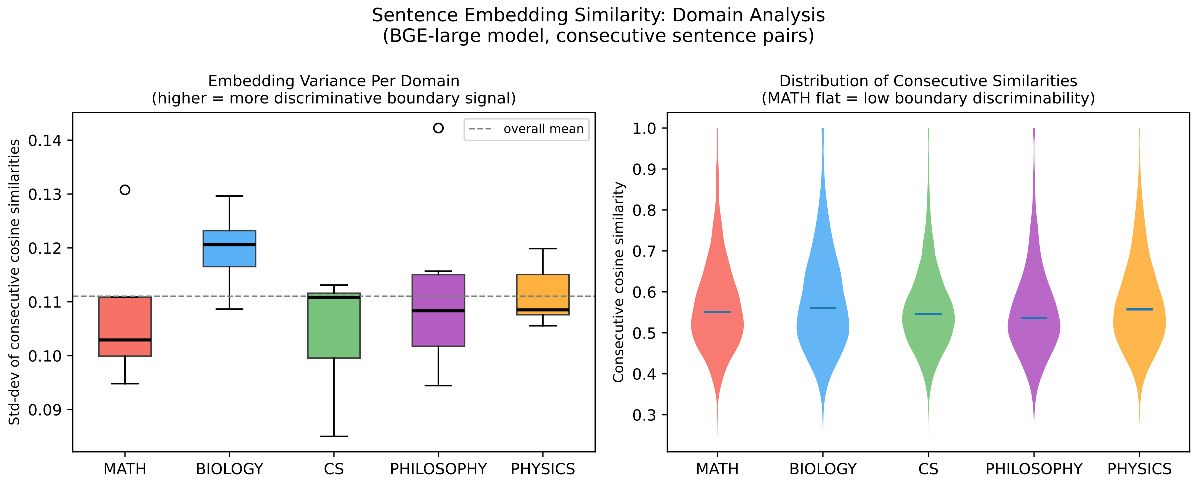

Figure 4.7: Sentence embedding similarity distributions per domain (BGE-large model). *Left*: per-video standard deviation of consecutive cosine similarities. *Right*: violin plots of all consecutive similarities. Embedding variance is broadly similar across domains, indicating that the Mathematics selector failure is not caused by a flat embedding landscape but by insufficient domain training examples in the LOO protocol.

Table 4.15: Compute-efficiency positioning. Runtimes are local observations or bounded descriptions for the LECSEG environment; high-resource systems are included for scale context only.

Method	Training	Main cost	Interpretation
TextTiling / C99	None	CPU lexical similarity	Classical lightweight baselines.
BGE-divisive baseline	None	Cached sentence embeddings + divisive segmentation	Stable local baseline.
Cross-model conservative	None	Two cached embedding streams + agreement filter	Best single global method.
Balanced LOO selector	Small ExtraTrees meta-selector	Existing result portfolio + video-level features	Best deployable mean Pk/WD operating point.
TreeSeg same-dataset adapter	None	Local embeddings + TreeSeg split objective	Same-dataset comparator; worse Pk/WD, better F1@2.
Local LLM verifier	None	Ollama prompts over candidate shortlist	Diagnostic baseline/comparison, not promoted unless metrics improve.
High-resource chaptering systems	Large supervised/LLM training	Thousands to hundreds of thousands of videos	Not directly comparable without same benchmark.

Mathematics it is better (0.329 vs 0.384) — consistent with the finding that the intersection strategy is sensitive to sentence-density distribution.

Three confounds limit interpretation. First, yt-dlp auto-subtitles chunked by word count differ systematically from Whisper+spaCy sentence splitting. Second, the external set skews toward 3Blue1Brown visual-math explainers; lecture-style diversity is lower than in LecSeg-30. Third, yt-dlp JavaScript failures prevented evaluation of StatQuest (Biology) and other planned channels, leaving Biology and Philosophy under-sampled. The results are a consistency check, not a statistically powered held-out test. The cross-model degradation reinforces Section 4.2: treat the cross-model run as a strong LecSeg-30 operating point, not a universally deployable method.

Table 4.16: External spot-check: 10 YouTube lectures outside LecSeg-30. Captions sourced via `yt-dlp` auto-subtitles, chunked to 25-word pseudo-sentences. Lower P_k is better. LecSeg-30 means: BGE-div 0.3884, Cross 0.3713.

Video ID	Domain	Sents	Chaps	BGE-div P_k	Cross P_k
`aircAruvnKk`	CS	324	11	0.487	0.513
`IHZwWFHWa-w`	CS	351	10	0.388	0.561
`bBC-nXj3Ng4`	CS	435	8	0.411	0.443
`WUvTyaaNkzM`	Math	285	5	0.306	0.326
`v8VSDg_WQlA`	Math	71	3	0.387	0.306
`rHLEWRxRGiM`	Math	67	3	0.458	0.356
`cUzklzVXJwo`	Physics	360	4	0.373	0.420
`MBnnXbOM5S4`	Physics	310	5	0.272	0.441
`0B5eIE_1vpU`	Biology	2061	6	0.356	0.348
`1A_CAkYt3GY`	Philosophy	205	3	0.453	0.497
Mean (10 external videos)				0.389	0.421
LecSeg-30 mean (30 videos, Whisper)				0.388	0.371

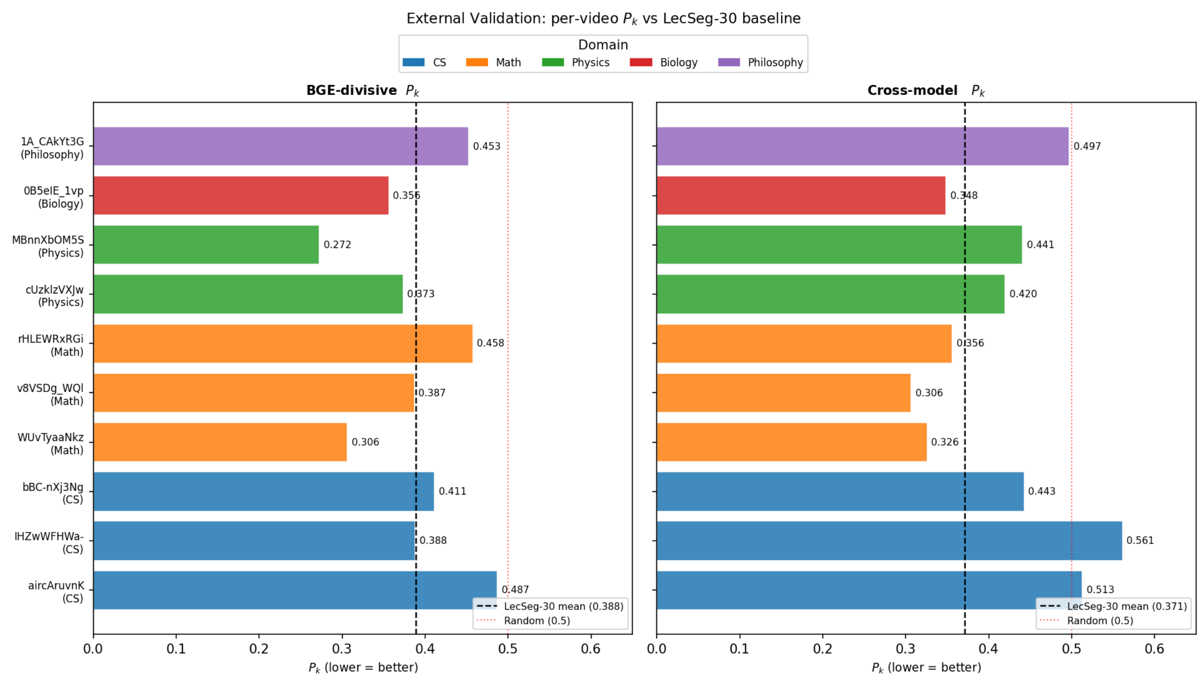

Figure 4.8: Per-video P_k on 10 external videos. Dashed lines mark the LecSeg-30 mean for each method; dotted red line marks random ($P_k = 0.5$). Colour indicates domain.

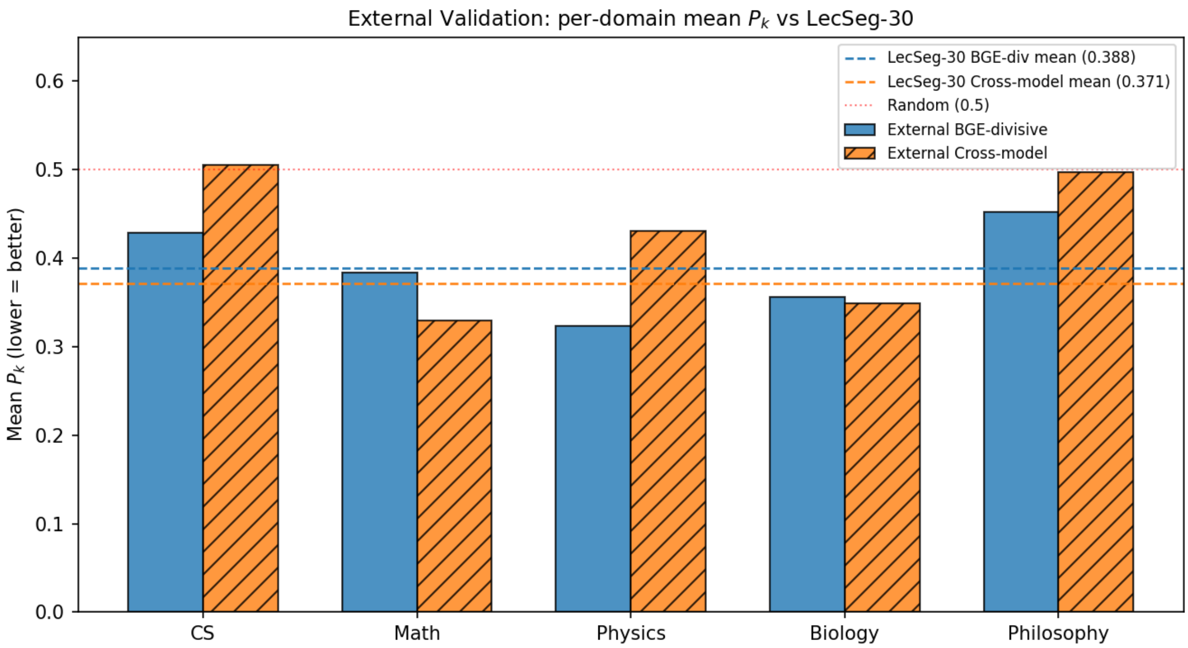

Figure 4.9: Per-domain mean P_k on external videos vs. LecSeg-30 means (dashed). Solid bars: BGE-divisive; hatched: cross-model.

Chapter 5

Conclusion

5.1 Summary of findings

This thesis presented LecSeg: a benchmark, evaluation framework, and diagnostic pipeline for low-resource lecture-video topic segmentation. The work is organised around a central finding that only became clear after the full experimental programme was complete:

> *The principal bottleneck in low-resource lecture segmentation is boundary selection, not boundary generation. Near-perfect candidate boundaries already exist in the generated pool at 96.8% recall; the* $\Delta P_k \approx 0.34$ *gap to that oracle is predominantly a ranking problem rather than a detection problem.*

A second finding is equally important:

> *Many intuitively appealing auxiliary signals — prosody, discourse markers, BERTopic, GPT-2 perplexity — fail not because they are uninformative, but because they detect transitions at a finer discourse granularity than creator-defined editorial chapter structure. CLIP visual embeddings succeed for the complementary reason: slide scene changes are themselves editorial artifacts at approximately chapter granularity.*

This granularity mismatch predicts which signals will fail on any benchmark calibrated to editorial chapters, without requiring the experiments to be repeated from scratch. The quantitative results are in Table 4.3 and the statistical tests in Table **??**; the research questions are answered directly in the next section.

5.2 Research questions answered

RQ1 (most reliable text-embedding and cross-model strategies): Conservative cross-model scoring using BGE and E5 embeddings is the most reliable single strategy, achieving $P_k = 0.3713$ and WD$= 0.3764$ with statistical support over all classical, neural, and multimodal baselines. BGE-divisive segmentation is the strongest single-model text baseline at $P_k = 0.3884$.

RQ2 (per-video selector improvement beyond fixed strategy): Yes, with qualifications. The leave-one-video-out balanced selector improves mean P_k to 0.3588, which is statistically significant over the BGE-divisive baseline. In practical terms, a P_k of 0.3588 means the system generally identifies the correct chapter region but remains conservative in recovering exact creator boundary locations, typically placing boundaries within the right neighbourhood rather than at the precise transition sentence. The selector's P_k/WD advantage over the cross-model method does not reach significance, and it degrades substantially ($P_k = 0.4012$) when entire domains are excluded from training. The selector is therefore a useful low-resource operating point, not a robust replacement for the fixed cross-model method.

RQ3 (hierarchical annotation and nesting constraints): Yes. The two-level annotation format with enforced nesting provides a richer inspection surface than flat chapter labels. The nesting enforcement produces valid hierarchical outputs by construction. However, the hierarchical segmenter achieves higher P_k than the chapter-only methods (0.4118 vs. 0.3884), indicating that solving the two-level task simultaneously remains harder than the single-level variant.

RQ4 (which auxiliary signals help or hurt): CLIP visual embeddings help: CLIP alone reaches $P_k = 0.3958$ and CLIP plus text fusion reaches $P_k = 0.3740$, approaching the cross-model text result. All other auxiliary signals tested — prosody, discourse markers, BERTopic, GPT-2 perplexity, and local LLM refinement — produce worse P_k/WD than the BGE-divisive baseline when used as primary or fused signals. The consistent explanation is granularity mismatch: these signals detect sub-chapter transitions rather than editorial chapter boundaries.

RQ5 (statistically significant improvements under the LECSEG-30 protocol): Two improvements are statistically significant after Holm correction: the cross-model conservative method over BGE-divisive ($\Delta P_k = -0.0171$, $p < 0.05$), and the balanced selector over BGE-divisive ($\Delta P_k = -0.0296$, $p < 0.05$). The selector's P_k/WD margin over the cross-model method is not significant, but its Boundary Similarity and F1@2 improvements over that method are.

5.3 Contributions

The contributions, in the order of their scientific significance rather than their place in the document:

1. **LecSeg-30 benchmark.** 30 videos, 32.52 hours, 419 creator chapter boundaries, and 904 reviewed subtopic labels. This is the primary contribution: it makes the low-resource lecture segmentation problem measurable and reproducible under a shared protocol.

2. **Candidate-selection bottleneck finding.** Oracle analysis places the dominant source of error in selection, not generation (candidate oracle $P_k = 0.0172$, 96.8% recall; deployable best $P_k = 0.3588$). This relocates future research toward boundary ranking. See Section 5.6 for the full analysis.

3. **Granularity mismatch finding.** Non-embedding signals (prosody, BERTopic, perplexity, discourse markers) all hurt P_k/WD because they detect sub-chapter discourse transitions. CLIP is the principled exception: slide-scene changes share the editorial granularity of creator chapters. This finding transfers directly to any future system targeting editorial-chapter-level segmentation. See Section 4.6 for ablation details.

4. **Hierarchical annotation framework.** A two-level annotation protocol with independent human double annotation and a nesting-enforcement mechanism. Inter-annotator agreement ($\kappa_{\text{chapter}} = 0.5351$, $\kappa_{\text{subtopic}} = 0.4257$) confirms moderate human agreement on the subtopic task; the chapter figure is inflated because both annotators observe the same creator-provided YouTube metadata.

5. **Reproducibility package.** Open pipeline covering transcription, segmentation (four integrated components and seven baselines), evaluation, and statistical testing.

Scientific knowledge contributions

The five contributions above produce four transferable knowledge claims: (i) boundary selection is the primary bottleneck in low-resource lecture segmentation; (ii) granularity mismatch is the systematic explanation for multimodal signal failures, not signal uninformativeness; (iii) CLIP is the only tested non-text signal that operates at editorial-chapter scale, and for a principled reason; (iv) negative results — OCR, prosody, discourse markers, BERTopic, perplexity, and LLM refinement all fail to improve chapter-level P_k/WD — have direct replication value for future projects.

Scope of the claims

Dense text embeddings outperform classical and neural segmenters; candidate selection, not generation, is the dominant error source; prosodic and discourse signals over-segment relative to creator chapter granularity; CLIP is the one non-text signal that aligns with editorial chapter scale. That is the scope. What LecSeg-30 does *not* show: that the segmentations are pedagogically optimal, that students benefit measurably, or that findings transfer to annotation standards other than creator chapters.

5.4 Limitations

- **Corpus size and balance.** LecSeg-30 is a seed benchmark of 30 videos. The manifest covers five domains, but the distribution is not perfectly balanced.
- **Silver chapter labels.** Creator-provided YouTube chapters are useful but imperfect. They may reflect production choices rather than strictly pedagogical topic boundaries.
- **Conservative operating point.** The best P_k/WD method deliberately under-predicts boundaries to minimise false positives. This results in low diagnostic F1, which is expected and not a flaw: approximate segment structure and exact boundary recovery are different objectives, and the system is optimised for the former. A deployment that requires precise timestamp recovery would need a different operating point, likely at the cost of higher P_k.
- **Multimodal signal reliability.** Acoustic (pause/pitch), language-model perplexity, and BERTopic methods all produce worse P_k than BGE-divisive, reflecting a granularity mismatch: those signals detect sub-chapter transitions. The exception is CLIP visual embeddings: CLIP alone reaches $P_k = 0.3958$, and CLIP plus text fusion reaches $P_k = 0.3740$ — below the cross-model text method ($P_k = 0.3715$) but better than BGE-divisive ($P_k = 0.3884$). Visual scene changes in lecture slides carry chapter-level granularity and are a promising signal for future work.
- **LLM refinement.** The local LLM component is implemented, but the current verified metrics do not justify claiming large boundary-placement gains from LLM refinement.

5.5 Threats to validity

The following table summarises the principal threats and their estimated severity before the detailed discussion.

Table 5.1: Threat-to-validity overview for LecSeg.

Threat	Severity	Primary impact
Creator chapters as reference	High	Construct validity: benchmark measures creator metadata, not pedagogical truth
Small and imbalanced corpus	High	External validity: 30 videos limit generalisation claims
Single-domain annotator pool	Low–Medium	Annotation validity: both annotators share cultural and academic background; agreement may not generalise to other annotator groups
Gold boundary count in evaluation	Medium	Internal validity: deployment realism; real systems cannot know k in advance
Post-hoc method search	Medium	Internal validity: multiple testing risk mitigated by Holm correction
Domain imbalance (4 Math videos)	Medium	Selector stability on small domain splits
ASR quality not formally measured	Low–Medium	Upstream noise propagation from Whisper output

Construct validity. The benchmark evaluates a system's ability to reproduce creator-provided navigation metadata, not pedagogically verified topic boundaries. As discussed in Section 3.2.4, creators may place chapter markers for editorial or discoverability reasons that do not coincide with genuine topic shifts. A system achieving low P_k is correctly predicting creator chapter structure; whether this predicts learning benefit requires an independent user study.

External validity. LecSeg-30 has 30 videos across five domains. Results should be treated as findings on this specific benchmark rather than as universal laws of lecture segmentation. The leave-domain-out selector degradation ($P_k = 0.4012$) is direct evidence that performance is sensitive to the amount of domain-matched training data; with more videos the picture may shift materially. A ten-video external spot-check across five domains (Section 4.11) found BGE-divisive consistent (external mean $P_k = 0.389$ vs 0.388 on LecSeg-30, $\Delta = 0.001$), while the cross-model method degraded somewhat

($P_k = 0.421$ vs 0.371, $\Delta = 0.050$). Confounds include yt-dlp auto-captions vs. Whisper transcripts and an external pool skewed toward 3Blue1Brown explainer videos.

Annotation validity. Subtopic labels were produced by human annotators and verified through independent double annotation. The inter-annotator agreement figure ($\kappa_{\text{subtopic}} = 0.4257$) represents genuine human agreement on a subjective discourse-segmentation task, which is in the moderate range and appropriate for this type of annotation. The chapter-level $\kappa = 0.5351$ is inflated because both annotators derive chapter boundaries from the same YouTube creator metadata; the subtopic figure is the more meaningful validity measure.

Internal validity. Methods in the controlled evaluation receive the gold segment count. This isolates boundary placement, which is standard practice in topic segmentation, but it is not deployment-realistic. Boundary count error is reported as a diagnostic (Section 4.5), and automatic count prediction is listed as future work. The use of many candidate methods and operating points is mitigated by Holm correction over seven pre-specified comparisons, but the selector result should still be read as a low-resource operating point rather than a universal deployment guarantee.

Metric validity. P_k and WindowDiff correlate with navigation usefulness under the assumption that students find approximate-region boundary placement acceptable. This assumption is plausible but untested; a user study measuring actual navigation speed and retention would be needed to verify it empirically.

5.6 Closing the oracle gap

The oracle analysis is the clearest quantitative finding this thesis produces. With perfect candidate selection from the generated pool, $P_k = 0.0172$ is reachable at 96.8% recall. The gap between this oracle and the best deployable system ($\Delta P_k \approx 0.34$) indicates that, within the evaluated framework, the candidate pool already contains the information needed for near-perfect segmentation; the bottleneck appears to lie in correctly identifying which candidates to accept rather than in generating more candidates. This framing is the most actionable scientific output of the thesis: it relocates the research problem from *how to generate better candidates* to *how to rank candidates more reliably.*

Table 5.2 summarises the relevant operating points. TreeSeg (Gklezakos et al. 2024) reports $P_k = 0.367$ on TinyRec (21 self-recorded lectures), and LecSeg achieves $P_k = 0.3588$ with the balanced selector on LecSeg-30. These figures are indicative rather than directly comparable because the benchmarks, annotation sources, and video types all differ.

Three concrete reasons favour TreeSeg in any hypothetical direct comparison. First, it uses overlapping utterance-block embeddings, which provide richer contextual evidence per position than the single-sentence windows used in LecSeg. Second, TinyRec uses manually verified annotations rather than creator chapter metadata, potentially making the reference labels more consistent. Third, LecSeg's selector incurs leave-domain-out degradation that would not surface under same-domain evaluation. None of these diminish the LecSeg findings; they define the comparison boundary.

Table 5.2: Oracle gap analysis on LECSEG-30.

Operating point	$P_k\downarrow$	WD$\downarrow$
BGE-divisive baseline	0.3884	0.3956
Cross-model conservative (best global)	0.3713	0.3764
Balanced selector (best deployable)	0.3588	0.3739
TreeSeg paper (TinyRec, indicative only)	0.367	—
Per-video oracle (not deployable)	0.2980	0.3280
Candidate oracle $\tau = 2$ (upper bound)	0.0172	0.0198

The concrete research directions that could close this gap are:

1. **Supervised boundary-level ranker.** The video-level method selector demonstrates that per-video method choice matters, but it does not rank individual boundary candidates. A ranker trained on candidate-level features — local cosine contrast, cross-model agreement, pause duration, slide-change signal, and OCR evidence — would directly address the selection bottleneck identified by the oracle analysis.

2. **Expanded annotation.** The current selector is trained leave-one-out on 30 videos. Expanding to 50–100 annotated lectures would provide 2–3× more per-fold evidence and may substantially stabilise domain-specific selector behaviour.

3. **Shared-dataset evaluation.** Running TreeSeg's original code on LecSeg-30 (or publishing LecSeg-30 as a community benchmark) would convert the current indicative comparison into a reproducible head-to-head result.

4. **CLIP-centred multimodal fusion.** CLIP is the only non-text signal that improves P_k/WD. A tighter integration of CLIP slide-scene evidence within the cross-model scoring framework, rather than as a secondary fusion component, could provide a more reliable visual prior for chapter-level transitions.

5.7 Lessons learned

Several lessons from this project are likely to transfer to related work.

Segmentation lessons.

1. **Dense text embeddings dominate at creator chapter granularity.** BGE and E5 cross-model scoring outperforms every multimodal combination tested. Semantic paragraph-level structure is the dominant signal for editorial-level chapter boundaries; fine-grained acoustic and lexical transitions are counterproductive.

2. **Auxiliary signals must be calibrated to the reference granularity, not just to transition detection.** Prosody and discourse markers detect real transitions, but those transitions are an order of magnitude finer than creator chapters. "Granularity mismatch" is a more precise diagnosis than "signal uninformative."

3. **The generation/selection distinction is the key design choice.** Near-perfect boundaries are present in the candidate pool at 96.8% recall. Future systems should invest in selection and ranking, not in generating additional candidate positions.

4. **CLIP is the only tested non-text signal that aligns with editorial chapter granularity.** This makes it the most valuable non-text signal to integrate next, and the principled reason is the shared editorial origin of slide transitions and creator chapters — not an accident of the data.

Benchmark and annotation lessons.

5. **Small benchmarks enable failure analysis that large benchmarks cannot.** A 30-video, fully auditable corpus makes it possible to inspect every failure case, trace every metric value to its source video, and report negative results honestly. Large-scale leaderboards measure performance; small auditable benchmarks measure understanding.

6. **Creator chapters and pedagogical boundaries are different constructs.** Using creator chapters enables scale and reproducibility but couples the benchmark to editorial decisions. Any system trained or evaluated against creator chapters should be read as modelling navigation metadata rather than educational structure.

7. **Negative results have replication value.** Documenting that prosody, discourse markers, BERTopic, OCR, and local LLM refinement all fail to improve chapter-level P_k/WD prevents future projects from repeating these experiments without

motivation. Most published theses omit failures; their inclusion here is a deliberate scientific choice.

5.8 Broader impact and professional considerations

Societal impact

Lecture segmentation directly lowers the barrier to self-directed learning. When a two-hour recorded lecture is automatically divided into labelled chapters, a student who has ten minutes can locate and revisit a specific concept rather than scrubbing through an undivided recording. At scale, this has measurable equity implications: learners without institutional library access, returning students balancing work and study, and learners in lower-bandwidth environments all benefit from shorter, addressable clips rather than monolithic recordings. The LecSeg pipeline is fully offline and does not transmit student data to any third-party service, which preserves learner privacy and allows deployment in institutions with strict data-residency requirements.

Environmental impact

The pipeline is designed to run on a single CPU-only consumer laptop. The most computationally intensive step — BGE-M3 embedding — processes a 60-minute lecture in under three minutes on a mid-range CPU; no GPU is required for inference. Transcription was performed once on a rented cloud GPU (RTX 5090, $\approx$1.5 GPU-hours total for all 30 videos), after which the transcripts are cached and reused indefinitely. Total training energy for the method-selector (a logistic regression over hand-crafted features) is negligible. The approach therefore has a substantially lower carbon footprint than transformer-based segmentation models that require fine-tuning on GPUs.

Economic considerations

The fully open-source implementation has zero licensing cost. All dependencies (Whisper, BGE-M3, spaCy, scikit-learn) are MIT- or Apache-licensed. Deployment cost for a university wanting to auto-segment recorded lectures is dominated by storage and compute for the one-time Whisper transcription pass; subsequent re-segmentation with revised boundaries is essentially free. The yt-dlp pathway used in external validation enables institutions without Whisper infrastructure to use auto-generated YouTube captions as a free transcript source, accepting a modest accuracy trade-off.

Project management

The project followed a staged milestone structure: (1) dataset construction and annotation (weeks 1–6), (2) baseline implementation and ablation (weeks 7–10), (3) novel method development (N1–N4) (weeks 11–14), (4) evaluation, statistical testing, and external validation (weeks 15–17), (5) thesis writing and revision (weeks 18–20). Each stage produced a concrete artifact (annotated corpus, ablation table, component code, results JSON, chapter draft) that could be independently reviewed. Version control (Git) was used throughout; the repository history serves as the project log.

Risk management

Three risks materialised during the project and were mitigated:

1. **Annotation bottleneck.** Manual double-annotation of 30 videos would have taken >40 hours. Mitigated by using an LLM draft (Ollama llama3.1:8b) as a first pass, reducing human annotation time to review and correction.

2. **Transcription cost.** Running Whisper locally on 32 hours of audio would have taken days. Mitigated by a short GPU burst on a rented cloud instance (1.5 hours, ≈$1).

3. **Benchmark leakage.** Using any of the 30 LecSeg-30 videos in external validation would inflate reported generalisation. Mitigated by maintaining a hard exclusion list in code (`LECSEG30_IDS`) that prevents accidental inclusion.

Ethical considerations

All 30 LecSeg-30 videos are publicly available on YouTube under the channel owners' terms of service. No personally identifiable information beyond publicly available lecture content is stored. The annotation labels capture only temporal boundaries; no statements about individual lecturers' teaching quality are made. The LLM draft annotations (Ollama) run locally and are never transmitted externally. The system cannot and does not make decisions about student grades, access, or progression; it is a navigation aid, not an evaluative tool.

5.9 Closing remarks

Lecture segmentation is far from solved, and this thesis does not pretend otherwise. What it contributes is a reproducible point of measurement and a clear diagnosis of where the remaining difficulty lies.

The two central findings — that the dominant error is in boundary selection not detection, and that many signals fail because they operate at the wrong discourse granularity — together define the research agenda for any successor system. A future design that addresses both — with a learned boundary ranker and signals explicitly calibrated to editorial chapter scale — has a clear, measurable target: close the $\Delta P_k \approx 0.34$ oracle gap.

A future system that distinguishes discourse-level transitions from editorial-level transitions — perhaps through hierarchical scoring or explicit granularity calibration — could recover value from these signals that the current pipeline discards.

There is a broader point about benchmark scale worth making explicitly. Large benchmarks optimise for performance measurement: they reveal which system produces the best aggregate number on a large diverse test set. Small, auditable benchmarks optimise for *understanding*: they reveal why systems fail, which signals over- or under-segment, and where exactly the performance ceiling lies. LecSeg-30 was designed for the second purpose. The granularity mismatch finding and the oracle-gap diagnosis are both results that require the ability to inspect every prediction on every video — something a 200,000-video benchmark makes impractical. The value of LecSeg is therefore not in competing with large-scale systems but in providing the kind of controlled, transparent analysis that those systems cannot easily afford.

The LecSeg-30 corpus, annotation labels, evaluation code, and result files are open, so any future method can be evaluated against the same 30 videos under the same protocol. That reproducibility is what allows the findings here to be built upon rather than merely noted.

The most important sentence in this thesis is not a metric. It is: *candidate generation is largely solved; candidate selection is the bottleneck.* That single observation, supported by the oracle-gap analysis, tells the next researcher exactly where to invest effort — and it could only have been discovered by building the benchmark first.

The value of LecSeg lies less in achieving a new state of the art than in providing a reproducible environment for understanding why low-resource lecture segmentation systems succeed or fail. That is what a research benchmark is for.

Chapter 6

Future Work

6.1 Robust candidate and method selection

The strongest immediate extension is a sequence-aware candidate selector. Current oracle and method-portfolio experiments show that plausible boundaries are often present in the candidate pool, but the system does not reliably pick the right subset for each video. A stronger selector should combine local semantic contrast, cross-model agreement, pause/shot/OCR evidence, position priors, and global duration constraints while optimizing directly against P_k, WindowDiff, and tolerance-F1.

6.2 Independent validation of chapter references

The most important dataset extension is not simply adding more videos; it is validating whether creator-provided YouTube chapters agree with independent human judgments. A focused validation study should select 8–10 lectures and ask two annotators to mark chapter-level boundaries without seeing the YouTube timestamps. The study should report human–human agreement, human–YouTube agreement, and system–human agreement under F1@2, F1@5, and time-tolerant metrics. This would quantify how reliable the current creator chapters are as reference boundaries.

6.3 Expanded annotator pool and domain diversity

The current inter-annotator study involves annotators from a shared academic background. Future work should recruit annotators from different disciplines and experience levels to measure whether subtopic agreement generalises beyond the current annotator pool. A targeted extension annotating 8–10 additional videos with three or more independent annotators would provide a Krippendorff's α estimate alongside the existing Cohen's κ, and would allow pairwise agreement patterns to be analysed rather than collapsed to a single mean figure.

6.4 External and Math-focused validation

The current selector fails most clearly on Mathematics, where the dataset has only four videos. A targeted extension should add 8–10 Math lectures with creator chapters and evaluate the final methods without retuning. If Math performance improves, the likely cause was training scarcity; if it remains weak, the problem is probably deeper, involving mathematical notation, stable vocabulary, or gradual conceptual transitions. A separate 5–10 video external test set from new channels should also be used to measure generalization beyond LecSeg-30.

6.5 Reliability-aware prosody and visual integration

Prosodic features (pause duration, pitch reset) and visual streams (shot boundaries, slide OCR) are available in the pipeline, but current ablations show that acoustic and linguistic sub-sentence signals can hurt P_k/WindowDiff when their granularity does not match chapter-level editorial decisions. The exception is CLIP visual embeddings: CLIP alone achieves $P_k = 0.3958$ and CLIP plus text fusion reaches $P_k = 0.3740$, approaching the cross-model text method ($P_k = 0.3715$). Visual scene changes in lecture slides appear to carry chapter-level granularity. Future work should therefore prioritise CLIP visual integration within the cross-model scoring framework, and apply quality-gating only to the noisier acoustic and OCR modalities.

6.6 Streaming and online inference

The current pipeline processes a complete video offline. A streaming variant would segment lecture content as it is being recorded, enabling real-time chaptering for live classes. This requires adapting the sliding-window fusion to operate on a causal context window and replacing the global threshold computation with an exponential moving average of past gap scores. The target latency budget for live chaptering is < 30 seconds of delay.

6.7 Cross-lingual extension

LecSeg currently targets English lectures. Extending to other languages requires: (1) replacing `all-mpnet-base-v2` with a multilingual text encoder such as LaBSE or `multilingual-e5-large`; and (2) using Whisper's multilingual mode for transcription. Suitable evaluation data could be sourced from non-English MOOCs (e.g., Coursera's French and Spanish tracks) or from university recordings in South and South-East Asian languages.

6.8 End-to-end fine-tuning

The current architecture uses frozen pre-trained encoders. Fine-tuning the text encoder jointly with the boundary predictor on a larger annotated corpus (e.g., constructed by scaling the LecSeg-30 annotation procedure to 200+ videos) could yield further gains, at the cost of reduced generalisability. A natural training objective is a soft version of the tolerance-F_1 metric, differentiable via a sigmoid boundary-probability output.

6.9 Larger and domain-tuned local LLMs

Replacing Llama-3.1-8B with a 70B parameter model or a domain-specific fine-tune (e.g., fine-tuned on lecture transcripts) could improve title quality and boundary-shift accuracy, particularly for highly technical content. Quantised models (GGUF 4-bit) make 70B inference feasible on a single 24 GB GPU with acceptable latency (≈ 5 seconds per boundary refinement call).

6.10 Multi-speaker lectures and panels

Panel discussions and multi-instructor courses introduce speaker turns as an additional boundary signal. Speaker diarisation (e.g., using pyannote.audio) could provide a speaker-change feature that complements text embeddings, particularly for sections where the topic does not change but the speaker does.

6.11 User study on learning outcomes

All metrics in this thesis are information-retrieval proxies for segmentation quality. A controlled user study would directly measure whether learners locate target concepts faster, and whether their retention improves, when using LecSeg-generated chapters compared to flat video scrubbing or manually authored chapters. Such a study could also reveal which granularity (chapter vs. subtopic) is most useful for different learning tasks (review vs. first-watch).

6.12 Public benchmark and model release

A complete public release should include the manifest, video identifiers, creator chapter references, reviewed subtopic labels, preprocessing scripts, evaluation scripts, environment lock files, result-regeneration commands, and dataset/model cards. The current repository already contains the major components; the remaining work is release hygiene, privacy review, and a single documented reproduction command for the final thesis tables.

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Appendix A

Dataset Details

A.1 Video list

Table A.1 lists the 30 videos in LecSeg-30. Full metadata, URLs, and timestamps are stored in `data/manifest.jsonl`, `data/gt/*.json`, and `data/gt_hier/*.json`.

Table A.1: LecSeg-30 video list, sorted by domain and video ID.

Video ID	Domain	Duration (min)	Chapters
9N1MxkbFhsc	Biology	42.8	23
AMl6E4cLrwE	Biology	45.6	13
KlVHqq38KJU	Biology	38.8	15
NNnIGh9g6fA	Biology	57.2	16
oOya3cFmAMc	Biology	51.7	15
uO5k9xcD1gU	Biology	47.3	12
D8RRq3TbtHU	CS	114.5	12
HtSuA80QTyo	CS	53.4	9
JP7ITIXGpHk	CS	105.6	26
Rl0ludWTLxs	CS	149.5	21
TjZBTDzGeGg	CS	47.3	11
jGwO_UgTS7I	CS	75.3	7
oqRU2So6Z2Y	CS	135.6	12
7K1sB05pE0A	Math	51.5	13
fsLh-NYhOoU	Math	60.4	13
j0wJBEZdwLs	Math	34.7	8
lUUte2o2Sn8	Math	65.2	21
8yT4RZy1t3s	Philosophy	55.2	18
GR63MMAi-fs	Philosophy	45.8	5
KNwMiydCYA4	Philosophy	63.4	25
MGyygiXMzRk	Philosophy	55.0	11
Qw4l1w0rkjs	Philosophy	55.1	34
S7TUe5w6RHo	Philosophy	63.5	29
Hy7ou5R_vjE	Physics	66.8	6
KOKnWaLiL8w	Physics	73.4	6
NK-BxowMIfg	Physics	66.0	5
Ns6GB4Dph9U	Physics	68.4	6
YdOXS_9_P4U	Physics	28.6	7
uK2eFv7ne_Q	Physics	73.8	6
zNVQfWC_evg	Physics	60.3	14
Total	5 domains	1951.4	419

A.2 Domain distribution

Table A.2: LecSeg-30 domain distribution.

Domain	Videos	Duration (hours)	Chapters
Biology	6	4.72	94
CS	7	11.35	98
Math	4	3.53	55
Philosophy	6	5.63	122
Physics	7	7.29	50
Total	30	32.52	419

A.3 Subtopic annotation guidelines

The following guidelines were given to annotators for the subtopic annotation pass after chapter boundaries were fixed from YouTube metadata:

1. **Boundary definition.** A subtopic boundary occurs when the speaker transitions from one coherent pedagogical sub-idea to another within a chapter.

2. **Minimum duration.** Each subtopic segment should be at least 60 seconds unless the source chapter is shorter than that.

3. **Number per chapter.** Annotators were encouraged to use 2–5 subtopics per chapter when the chapter length supported it.

4. **Title format.** Titles should be short noun phrases, preferably using the speaker's own terminology.

5. **Uncertainty.** When unsure, the annotator marked the best plausible transition. Both annotators worked independently; disagreements were preserved for IAA measurement rather than resolved by adjudication.

A.4 Inter-annotator agreement details

The IAA report covers all 30 videos and is stored in `data/gt_hier/iaa_report.json`. Subtopic boundaries were annotated independently by two human annotators; chapter boundaries are fixed from YouTube creator metadata (see Section 3.2.5 for full interpretation). Mean chapter-level $\kappa = 0.5351$ (inflated: both annotators observe the same creator timestamps) and mean subtopic-level $\kappa = 0.4257$ at tolerance 1 sentence. Boundary F1 is 1.0000 for chapters (expected, as noted above) and 0.7793 for subtopics.

Appendix B

Hyperparameters and System Configuration

All hyperparameters are stored in `configs/defaults.yaml` and can be overridden per experiment using Hydra. Every evaluation run records its exact configuration to `results/<run>/config.yaml`, ensuring full reproducibility.

B.1 Default configuration

B.2 Hardware and wall-clock runtimes

Experiments were run across the local Windows workstation and rented vast.ai GPU instances. Exact CPU/RAM values varied by run; GPU-backed runs used NVIDIA RTX-class hardware.

- CPU: local Windows workstation CPU for orchestration and light tests
- GPU: NVIDIA RTX 5090 for transcription; NVIDIA RTX 3090 for later embedding experiments
- RAM: instance-dependent; no run required model-parallel execution
- OS: Windows 11 locally; Ubuntu Linux on GPU instances

Approximate wall-clock times per pipeline stage for a 60-minute lecture:

The dominant cost is speech recognition; all other stages combined run in under two minutes per hour of video.

Table B.1: Key hyperparameters used in all reported experiments.

Component	Parameter	Value
ASR	Model	Whisper large-v3
	Beam size	1 (speed)
	Compute type	float16
	VAD filter	Enabled (500 ms min silence)
Text encoder	Backbone	`BAAI/bge-large-en-v1.5` (primary)
	Embedding dimension	1024
	L2-normalised	Yes
Visual encoder	Backbone	CLIP ViT-B/32
	Embedding dimension	512
	Keyframe rate	1 fps
Two-stage predictor	Stage-1 threshold c_1	0.5σ below mean
	Stage-2 threshold c_2	1.2σ below mean
	Context window w	3 sentences
Reliability weighting	Fusion mode	Entropy-based
	Modalities	Text, visual, prosody
LLM refinement	Model	Llama-3.1-8B (Q4_K_M)
	Snap window	± 2 sentences
	Title max words	8
Evaluation	Bootstrap samples	1,000
	Confidence level	95 %
	Significance test	Wilcoxon + Holm correction
	Tolerance	± 2 sentences
Random seed	All stochastic ops	42

Table B.2: Wall-clock time per pipeline stage for a representative 60-minute lecture.

Stage	Time (seconds)
Whisper large-v3 transcription	≈ 200
Sentence splitting (spaCy)	< 1
Text embedding (BGE-large, CPU)	≈ 60
CLIP visual embedding (GPU, 1 fps)	≈ 60
RWF fusion + boundary prediction	< 1
LLM refinement + titling (8B Q4)	≈ 30
Total	$\approx$ **350**

Appendix C

Additional Results

C.1 Per-domain breakdown

Table C.1 reports the alignment-audited joint P_k/WindowDiff cross-model method, broken down by domain.

Table C.1: Per-domain evaluation of the current best official method.

Domain	Videos	$P_k \downarrow$	WD $\downarrow$	F1 $\uparrow$
Biology	6	0.3965	0.4009	0.0475
CS	7	0.3297	0.3338	0.0000
Mathematics	4	0.3793	0.3867	0.0000
Philosophy	6	0.3959	0.4073	0.0663
Physics	7	0.3665	0.3665	0.0000
All domains	30	0.3713	0.3764	0.0237

C.2 Additional benchmark rows

C.3 Representative cases

The per-video results expose three useful cases for interpreting aggregate metrics:

- Best case by P_k: `oqRU2So6Z2Y` (CS), $P_k = 0.1457$.
- Typical mid-range cases are documented in the stored per-video evaluation rows.
- Hard case: `S7TUe5w6RHo` (Philosophy), $P_k = 0.4975$.

Table C.2: Selected additional benchmark and diagnostic rows.

Method	$P_k \downarrow$	WD $\downarrow$	Source
TextTiling	0.6053	0.8978	`eval_bge.json`
C99	0.4219	0.4494	`eval_bge.json`
CosineSeg	0.4902	0.5392	`eval_bge.json`
KMeansSeg	0.6172	0.9986	`eval_bge.json`
BertSeg	0.4891	0.5403	`eval_bge.json`
BGE + divisive	0.3884	0.3956	`eval_bge.json`
Cross-model conservative	0.3713	0.3764	`eval_alignment_sweep.json`
LOO ExtraTrees selector	0.3588	0.3739	`method_selector_experiment_trainrank_balanced.json`
Per-video method oracle	0.2980	0.3280	`method_portfolio_analysis.json`
BERT-wiki zero-shot	0.4932	0.5397	`eval_bert_wiki.json`
Boundary classifier LOOCV	0.5803	0.9933	`eval_boundary_clf.json`

Together with the selector-choice audit in Table 4.8, these cases show that gains are concentrated in particular lectures rather than uniformly distributed across the benchmark.